Revisiting Generic Transforms for Committing Authenticated Encryption

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Abstract. Committing security for authenticated encryption (AE) captures the difficulty of constructing a distinct input tuple, including the key, that yields the same ciphertext. This notion is relatively new but has attracted significant attention due to its practical relevance. A promising direction is to design generic transforms that convert any AE scheme into a committing one. A common approach to generic transforms, initiated by the CTX transform (Chan and Rogaway, ESORICS 2022), is to add a hash function that uses part of the AE input/output, assuming the hash is ideal, i.e., a random oracle. Because the baseline AE is assumed to be secure in the standard model, this approach inherently mixes standard-model and idealized-model assumptions. We revisit this approach. We show that a number of state-of-the-art generic transforms relying on a mixed model (Chen and Karadžić, Eurocrypt 2025, and Bhattacharjee et al., ePrint 2024), proposed after CTX, are vulnerable once the hash function is instantiated, by presenting practical counterexamples enabling efficient attacks. Our attacks/counterexamples exploit the fact that the baseline AE may depend on the instantiation of the generic transform, whereas the opposite is not true for the principle of the generic transform. In most cases, the attacks are effective with any instantiation, and the baseline AEs in the attacks have a natural structure, such as Enc-then-MAC with a counter mode encryption. We also demonstrate how to rectify these damaged transforms with minimal algorithmic modifications, relying solely on the standard-model assumptions.

Keywords: Authenticated Encryption, Committing security, Generic Transform, Provable security, Idealized model, Random Oracle

1 Introduction

(Committing) authenticated encryption. Authenticated Encryption (AE) is a symmetric-key cryptographic function that combines encryption and message authentication [BN00, Rog02]. Traditional AE security notions, namely privacy and authenticity, treat the key as a secret. The recent studies [FOR17, GLR17, DGRW18, LGR21, IIM21, ADG+22] have shown that these notions do not necessarily capture the security requirements for AE expected in real applications.

These works introduce the notion of committing security, which captures the infeasibility of finding two distinct input tuples that yield the same ciphertext. The crucial difference from the traditional black-box notions is that the adversary chooses the keys as part of the input. Because this notion is new, most standard AE schemes, such as GCM [Dwo07], do not satisfy committing security. The idea of committing security for AE was first proposed by Farshim *et al.* [FOR17], followed by formalization and classification of various committing security notions [BH22, CR22]. Now, committing AE is a very active and timely research topic for its practical relevance [GLR17, DGRW18, IIM21, ADG+22, LGR21].

A promising approach to achieving a committing AE is to design generic transforms that convert any AE scheme into a committing one. Chan and Rogaway [CR22] introduced an elegant transform, CTX, which augments the baseline AE scheme Π with a single hash function H. CTX computes $T^* = \mathsf{H}(K, N, A, T)$, where K is the key of Π, N is a nonce, A is associated data, and T is the tag produced by encryption of Π. While the original output is (C, T) for the ciphertext C, the output of CTX is (C, T^*), hence it saves T from the output (see Fig. 12). Chan and Rogaway proved that CTX transforms any black-box secure Π into a context-committing AE, assuming H is modeled as a Random Oracle (RO). Here, context committing [BH22, CR22] is the strongest committing security notion.

Despite its simplicity and efficiency, CTX has several limitations. It requires that the encryption algorithm of Π be decomposable into separate (plain) encryption and tag-generation routines. AE schemes with this structure are called Tag-based AEs [CR22], which include GCM and OCB [KR11]. Chen and Karadžić [CK25] (CK25) noted that CTX is not compliant with the AE standard API; specifically, the decryption routine must be modified. CK25 and Bhattacharjee *et al.* [BBD24] (BBD24) further noted that CTX does not support misuse-resistant AEs [RS06] and other non-tag-based AEs, which is a practical drawback. In summary, CTX is not strictly generic.[3]

Both CK25 and BBD24 proposed generic transforms that address these limitations while retaining the core idea of CTX, namely, hashing a portion of AE's input/output. In particular, CK25 introduced five strictly generic transforms that offer specific usability advantages over existing approaches. Their security proofs follow the CTX methodology; they prove AE security of the transform assuming black-box AE security of Π and assuming that H is an RO. The proposals of BBD24 have strong structural similarities with CTX but incorporate mechanisms to maintain API compatibility while keeping small ciphertext overhead. Consequently, CK25 and BBD24 represent the current state of the art in strictly generic transforms.

Our contributions. In this paper, we revisit the "hash-as-ideal" approach introduced and popularized by CTX. Although RO abstraction is extremely useful, in a real construction, ROs must be instantiated by concrete hash functions such

[3] A strictly generic transform is compatible with standard AE API and requires no additional security property on the baseline AE beyond the black-box AE security.

as SHA-2 or SHA-3 [Dwo15]. Because a generic transform must work with any AE scheme, the baseline Π may also employ SHA-2 or SHA-3. This creates a setting that diverges from the assumptions made in the original proofs, yet is both practical and unavoidable, since SHA-2 and SHA-3 are widely used to construct standard-model-secure functions such as PRFs and nonce-based (authenticated) schemes [KBC97, Sar10, DHM+25, RVV15]. Based on this observation, we show that most of the transforms proposed by CK25 and BBD24 yield counterexamples. For each transform Ω, we construct an AE Π (a counterexample) that uses the same instantiation of the hash function H employed by Ω. The resulting Π is natural in structure and provably secure with a reasonable *standard model assumption* on H. However, once transformed into $\Omega[\Pi, \mathsf{H}]$, the scheme becomes vulnerable to simple privacy/authenticity attacks, and in some cases even plaintext recovery or key recovery attacks. These attacks demonstrate that a malicious party could embed a backdoor into a protocol that relies on black-box AE schemes combined with such generic transforms. We did not invalidate the original mixed-model proofs of these transforms, hence our counterexamples reveal a fundamental gap in their proofs and their instantiations. Table 1 summarizes our results. We have informed the authors of CK25 and BBD24 of our results. Both acknowledged the validity of our attacks, though their assessments of the attacks' practicality are not entirely consistent with ours.[4]

The root cause of our attacks lies in the use of idealized models, an issue recognized for decades [CGH04]. However, our attacks are based on a slightly different context from those of the previous "uninstantiable RO"-based ones. More specifically, the issue in our setting arises from the mixed use of standard-model and idealized-model assumptions. This mixture obscures the precise conditions under which the baseline AE Π and the hash function H can be instantiated securely. In contrast, generic transforms that rely solely on standard-model assumptions do not suffer from this problem. Guided by this principle, we repair the damaged transforms using minimal algorithmic modifications together with standard-model assumptions such as collision resistance and pseudorandomness of the underlying primitives. For most transforms, we apply a pseudorandom generator (PRG) to the secret key to derive two independent keys, one for AE and the other for the transform, a technique we call Key Separation (KS). The computation of KS is lightweight and can be cached, making its online cost negligible (see Table 2). With additional collision resistance assumptions on the PRG and the hash, our fixed versions are provably secure under black-box standard-model AE security notions and context-committing security. By construction, they are immune to the instantiation-sharing attacks we showed against CK25 and BBD24, while preserving desirable features of the original transforms, such as avoiding nonce-based rekeying. We remark that nonce-based rekeying is used by many committing AE designs, such as [BH22]. Notably, two of our fixed transforms (AEaH-KS and AEtH-KS) are the *first* transforms that are 1) strictly generic, 2) context committing, 3) free from nonce-based rekeying,

[4] The authors of BBD24 fully acknowledged practicality, while the authors of CK25 expressed some different views.

Table 1: Our attacks (counterexamples). $\boldsymbol{\times}_e$: the attack which works for a class of instantiations. $\boldsymbol{\times}$ denotes the attack which works for any instantiation. ($\boldsymbol{\times}$): the attack based on slightly artificial, yet provably secure instantiation. **Model** denotes the security model: M = Mixed, S = Standard. Gray cells denote non-existence of attacks for entirely relying on the standard model.

Scheme	Ref.	Generic	Nonce-based Rekey	Model	Our Attack Priv	Auth	Key Recov.
CTX	[CR22]	✗		M	–	–	–
CTY	[BH24]	✗		M	–	–	–
HtE-UtC/RtC	[BH22]	✓	✓	S			
HtAE	[CK25]	✓	✓	M	–	–	–
AEaH	[CK25]	✓		M	✗	✗	(✗)
EtH	[CK25]	✓		M	✗	✗	(✗)
AEtH	[CK25]	✓		M	✗	–	(✗)
chaSIV	[CK25]	✓	✓	M	$\boldsymbol{\times}_e$	–	($\boldsymbol{\times}_e$)
PACT	[BBD24]	$\approx$ ✓*		M	✗	✗	(✗)
comPACT	[BBD24]	$\approx$ ✓*		M	✗	✗	(✗)

∗: API is generic but requires a non-standard property for AE

and 4) standard-model secure. Although our transforms introduce little technical novelty, they are meaningful in light of these combined properties. Finally, we briefly analyze CTX and its refinement CTY by Bellare and Hoang [BH24]. We do not identify attacks against them despite their reliance on a mixed model, and we show that standard-model-secure variants can be obtained straightforwardly using the KS technique (see Sect. 5).

Related Work. Bellare and Rogaway [BR93] introduced the RO model and demonstrated its usefulness in provable security. Since the seminal work of Canetti, Goldreich, and Halevi [CGH04], uninstantiable constructions in the RO model have been extensively studied, including those by Bellare, Boldyreva, and Palacio [BBP04] and Brzuska, Farshim, and Mittelbach [BFM15]. Black [Bla06] presented an uninstantiable construction based on the ideal cipher model. More recently, Khovratovich, Rothblum and Soukhanov exploited the RO abstraction in the Fiat-Shamir transform [KRS25]. Committing attacks against widely deployed AE schemes have been reported in many works [GLR17, DGRW18, ADG+22, LGR21, CR22, MLGR23]. Bellare and Hoang presented a variety of (generic) transforms for committing AE, including CTY [BH24], and HtE, UtC, and RtC [BH22]. Transforms dedicated for Encode-then-Encipher (EtE) [BR00] style AE were proposed by [HM24b, NSS25]. Dedicated constructions based on popular AE and hash designs, such as GCM, CCM [Dwo04], Ascon [TMK+25], and SHA-3, have been proposed [BH22, NSS24, NSS23, DHM+25]. Krämer, Struck, and Weishäupl [KSW24] studied the committing security of the finalists of the NIST Lightweight Cryptography Project [NIS], showing positive results for those based on Sponge/Duplex [BDPV08, BDPV12]. Committing security of Duplex was ana-

Table 2: Our fixes for the CK25 generic transforms. **Alg. chg**: Algorithmic change. **Nb rekey**: Nonce-based rekey. **Add. comp**: Additional cryptographic computation, represented by the total number of increased/decreased input bits given to the primitives, where κ denotes the key bit length.

Scheme	Fix	Alg. chg	Nb Rekey	Add. comp
HtAE-ST	H as RCR-PRF	no	✓	0
AEaH-KS	Key Separation	yes		$O(\kappa)^{\dagger}$
EtH-KS	Key Separation	yes		$O(\kappa)^{\dagger}$
AEtH-KS	Key Separation	yes		$O(\kappa)^{\dagger}$
chaSIV-NKS	N-b Key Sep. + remove nonce from H	yes	✓	0^{*}

†: Computation can be cached per key.
∗: Use of a PRF of domain $\mathcal{N}$ (nonce space) reduces input of H by $\log|\mathcal{N}|$ bits.

lyzed by [DFG23]. Additional committing AE constructions have been proposed in works such as [FOR17, GLR17, DGRW18, ADG+22, HM24a, SW24, BCC+24].

Known/generic problem? Generally, reusing (instantiations of) idealized primitives is a known pitfall of provably secure systems. For example, Bellare et al. [BDG20] investigated NIST Post-quantum Key Encapsulation Mechanism (KEM) proposals which require multiple independent ROs and found insecure RO instantiations by the designers. The most typical one is reusing a single hash function without any modification. Similarly, our counterexamples in Section 4 may be viewed as attacks under "forbidden" ideal primitive sharing, e.g., sharing an RO, H, between the transform and the baseline AE, without considering instantiations. This view is practically valid, particularly given numerous AE proposals based on ideal primitives. However, it implies that the AE is ideal-primitive-based, thereby contradicting its standard-model-secure assumption. Moreover, our attack principle gives a possibly malicious AE designer more freedom, since the baseline AE in the transform is a black box, not specified with a primitive. For example, an attacker under the mixed model may use a modified version of (the instantiation of) H as a primitive, or even the baseline AE may be a dedicated construction, e.g., based on the round function of H.

The difference between our mixed-model attack scenario and the above shared-ideal-primitive scenario is subtle, and we think both are worth studying. We do not claim generic insecurity of mixed models. A mixed model would be a natural, logical choice for many applications, such as hybrid encryption involving, e.g., Fujisaki-Okamoto transform [FO99]. Still, our work suggests a potential, application-dependent risk of a mixed model.

Organization. In Sect. 2, we present preliminaries to define security notions and basic cryptographic primitives. Sect. 3 defines generic transform to committing AE and its security goals. In Sect. 4, we present our attacks against the instantiations of recently-proposed generic transforms by CK25 and BBD24. In Sect. 5, we present how to fix the CK25 transforms with concrete proposals and a brief analysis of CTX and CTY. Sect. 6 concludes the paper.

2 Preliminaries

Basic notations. For integers i and j with $0 \le i < j$, let $[i..j] \coloneqq \{i, i+1, \ldots, j\}$. Let $\{0,1\}^*$ be the set of all bit strings, including the empty string ε, the single element in $\{0,1\}^0$. For $X \in \{0,1\}^*$, $|X|$ denotes its bit length and $|X|_n$ denotes $\lceil |X|/n \rceil$. We define the parsing of a string into n-bit blocks for a positive integer n by $(X[1], X[2], \ldots, X[m]) \xleftarrow{n} X$, where $X[1] \,\|\, X[2] \,\|\, \ldots \,\|\, X[m] = X$, $|X[i]| = n$ for $1 \le i < m$ and $0 < |X[m]| \le n$ when $|X| > 0$. A concatenation of bit strings X and Y is denoted as $X \,\|\, Y$, or simply XY. Let 0^i be the string of i zero bits. Hence, we write 10^i for $1 \,\|\, 0^i$. For $X \in \{0,1\}^*$ with $|X| \ge i$, $\mathtt{msb}_i(X)$ is the first (left) i bits of X, and $\mathtt{lsb}_i(X)$ is the last (right) i bits of X. Let $\langle X \rangle_c$ be the c-bit standard representation of $X \in [0..2^c - 1]$. For example, $\langle 5 \rangle_4 = 0101$. If c is clear from the context, we may omit the subscript. If X is uniformly chosen from the set $\mathcal{X}$, we write $X \leftarrow\!\!\$\, \mathcal{X}$.

Oracles and advantage. Suppose $\mathcal{A}$ is an adversary in a game. $\mathcal{A}$ can query s oracles, $\mathcal{O}_1, \ldots, \mathcal{O}_s$, in any order. In the case of a distinguishing game, $\mathcal{A}$ outputs a bit after the queries, under the probability space defined by $\mathcal{A}$ and $\{\mathcal{O}_i\}_{i \in [1..s]}$. Let $[\mathcal{A}^{\mathcal{O}_1, \ldots, \mathcal{O}_s} = 1]$ denote the event that this bit is 1. We call $\mathcal{O}_i$ the i-th oracle in the game. By writing $\mathbf{Adv}^{\mathtt{xxx}}_{\Pi}(\mathcal{A})$, we mean the *advantage* of the adversary $\mathcal{A}$ in the game $\mathtt{xxx}$ involving Π, and $\mathcal{A}$ may be called $\mathtt{xxx}$-adversary. The time complexity of an adversary is denoted by $\mathtt{tm}$. By writing $\mathtt{tm}'$ we mean $\mathtt{tm}' = \mathtt{tm} + O(\theta)$ for some adversarial resource parameter θ specified in the game. We say Π is $\mathtt{xxx}$-secure if the corresponding advantage is negligible for all computationally bounded adversaries.

Primitives. Any keyed function $F : \mathcal{K} \times \mathcal{X} \to \mathcal{Y}$ with key space $\mathcal{K}$, we may write $F_K(x)$ to denote $F(K, x)$. Unless otherwise specified, we assume $K \leftarrow\!\!\$\, \mathcal{K}$ in evaluation of $F_K(x)$. A block cipher $E : \mathcal{K} \times \mathcal{M} \to \mathcal{M}$ is a keyed function such that for any $K \in \mathcal{K}$, $E(K, \cdot)$ is a permutation over the message space $\mathcal{M}$. We write $E_K^{-1}(\cdot)$ to denote the decryption function. Let $\mathrm{Func}(\mathcal{X}, \mathcal{Y})$ be the set of all functions: $\mathcal{X} \to \mathcal{Y}$. A uniform random function (URF) $\mathsf{R} : \mathcal{X} \to \mathcal{Y}$ is a random function uniformly chosen over $\mathrm{Func}(\mathcal{X}, \mathcal{Y})$.

An ideal (cryptographic) primitive, or a primitive under an idealized model, is a (not necessarily uniform) random function that models the ideal functionality of a cryptographic keyless/deterministic function. It is publicly accessible, that is, it has an internal randomness, typically exponentially large, and can be queried by any entity, including the cryptographic scheme built on it and an adversary against that scheme. An "instantiation" of an ideal primitive refers to the use of a concrete, keyless function instead of the ideal primitive. An instantiation means replacing a random function with a deterministic one; hence, it can never be correct. However, this has been used to prove the security of schemes for its extreme usefulness [BR93].

We describe typical ideal primitives. A random oracle (RO) is a public random function $\mathsf{RO} : \{0,1\}^* \to \{0,1\}^m$ for some fixed m. The output is uniformly random for each input. An ideal cipher (IC) is a public function $\mathsf{IC} : \mathcal{K} \times \mathcal{M} \to \mathcal{M}$ that

uniformly distributes over all the block ciphers with key space $\mathcal{K}$ and message space $\mathcal{M}$. If the key space of IC is a singleton, it reduces to a public random permutation (RP), RP. For being idealized primitive, RO, IC, and RP can be evaluated publicly, and IC/RP can be evaluated in both forward and backward directions. An RO idealizes cryptographic hash functions, and an IC idealizes block cipher encryption and decryption algorithms; note that the key is treated as a part of the input so that the target algorithm is keyless.

A primitive under a standard model is one that does not rely on an idealized model. We provide several definitions of such primitives.

PRG and PRF. For a function $\mathsf{F} : \mathcal{K} \times \mathcal{X} \to \mathcal{Y}$ and a URF $\mathsf{R}_\mathsf{F} : \mathcal{X} \to \mathcal{Y}$, let

$$\mathbf{Adv}^{\mathtt{prf}}_{\mathsf{F}}(\mathcal{A}) \coloneqq |\Pr[\mathcal{A}^{\mathsf{F}_K} = 1] - \Pr[\mathcal{A}^{\mathsf{R}_\mathsf{F}} = 1]|,$$

where $K \leftarrow\!\!\$\, \mathcal{K}$, denote the pseudorandom function (PRF) advantage of F against $\mathcal{A}$. If $\mathcal{A}$ is $(q, \sigma, \mathtt{tm})$-PRF adversary, it uses q queries with total σ blocks and $\mathtt{tm}$ time. When F takes fixed-length inputs, we omit σ. For a function $\mathsf{G} : \mathcal{K} \to \mathcal{K}'$,

$$\mathbf{Adv}^{\mathtt{prg}}_{\mathsf{G}}(\mathcal{A}) \coloneqq |\Pr[\mathcal{A}^{\mathsf{G}} = 1] - \Pr[\mathcal{A}^{\$} = 1]|,$$

where G returns $\mathsf{G}(K)$ for $K \leftarrow\!\!\$\, \mathcal{K}$ and $\$$ returns uniformly random sample on $\mathcal{K}'$, denote the pseudorandom generator (PRG) advantage of G against $\mathcal{A}$ (who has no query; only the response of the oracle is given). If $\mathcal{A}$ is a $\mathtt{tm}$-PRG adversary, it has $\mathtt{tm}$ time complexity.

(Right-half) Collision resistance. Let $\mathsf{H}_1 : \mathcal{X} \to \mathcal{Y}$ and $\mathsf{H}_2 : \mathcal{X} \to (\mathcal{Y}_1 \times \mathcal{Y}_2)$ be functions. For $(Y_1, Y_2) = \mathsf{H}_2(X)$, let $\mathsf{H}_2^{\mathrm{right}}(X) = Y_2$. We define Collision Resistance (CR) of H_1 against the adversary $\mathcal{A}_{\mathsf{H}_1}$ and Right-half Collision Resistance (RCR) [BH22] of H_2 against the adversary $\mathcal{A}_{\mathsf{H}_2}$ as

$$\mathbf{Adv}^{\mathtt{cr}}_{\mathsf{H}_1}(\mathcal{A}_{\mathsf{H}_1}) \coloneqq \Pr[(X, X') \leftarrow \mathcal{A}_{\mathsf{H}_1} : X \neq X', \mathsf{H}_1(X) = \mathsf{H}_1(X')],$$
$$\mathbf{Adv}^{\mathtt{rcr}}_{\mathsf{H}_2}(\mathcal{A}_{\mathsf{H}_2}) \coloneqq \Pr[(X, X') \leftarrow \mathcal{A}_{\mathsf{H}_2} : X \neq X', \mathsf{H}_2^{\mathrm{right}}(X) = \mathsf{H}_2^{\mathrm{right}}(X')].$$

If $\mathcal{A}_{\mathsf{H}_1}$ ($\mathcal{A}_{\mathsf{H}_2}$) is $\mathtt{tm}$-CR ($\mathtt{tm}$-RCR) adversary, it has $\mathtt{tm}$ time complexity. RCR is also introduced and used by [BH22, CK25].

Almost XOR universal hash function. A function $\mathsf{H} : \mathcal{K} \times \mathcal{X} \to \{0,1\}^n$ is said to be ϵ-almost XOR universal (ϵ-AXU) if

$$\max_{X, X' \in \mathcal{X}, X \neq X', \delta \in \{0,1\}^n} \Pr[\mathsf{H}(K, X) \oplus \mathsf{H}(K, X') = \delta] \leq \epsilon,$$

where K is uniformly random. A popular example of AXU hash function is GHASH inside GCM. It is a classical polynomial hash function defined over $\mathrm{GF}(2^n)$. For $X = (X[1] \,\|\, X[2] \,\|\, \ldots X[m])$ with $|X[i]| = n$, it is defined as $\mathsf{PH}(K, X) = \sum_{i \in [1..m]} K^{m-i+1} \cdot X[i]$ (additional length encoding may be needed for variable-length input). For the algebraic structure, once $\mathsf{GHASH}(K, (X_1, X_2))$ is known for some (X_1, X_2) with random K, we can learn K with a high probability. This property has been used by a classical nonce-reuse attack against GCM [Jou06].

2.1 Authenticated encryption

A nonce-based AE scheme $\Pi = (\Pi.\mathcal{E}, \Pi.\mathcal{D})$ is defined over a key space $\mathcal{K}$, a nonce space $\mathcal{N}$, an associated data (AD) space $\mathcal{AD}$, and a message space $\mathcal{M}$. An AD is a variable that is not encrypted, but its integrity must be ensured. Formally, AE is a tuple of an encryption function $\Pi.\mathcal{E}\colon \mathcal{K}\times\mathcal{N}\times\mathcal{AD}\times\mathcal{M}\to\mathcal{M}$, and a decryption function $\Pi.\mathcal{D}\colon \mathcal{K}\times\mathcal{N}\times\mathcal{AD}\times\mathcal{M}\to\mathcal{M}\cup\{\perp\}$, where symbol $\perp$ indicates a verification failure. We require $\Pi.\mathcal{D}_K(N,A,C) = M$ if $\Pi.\mathcal{E}_K(N,A,M) = C$ for correctness. For authenticity, a ciphertext must expand. If Π is a baseline AE scheme before transform (*i.e.*, not committing secure), we assume $|C| = |M| + \tau$ for some fixed, predetermined $\tau > 0$ for any encryption of (N,A,M). We also assume $\mathcal{K} = \{0,1\}^\kappa$. If Π is tidy, whenever $\Pi.\mathcal{D}_K(N,A,C) = M$ holds for some $M \neq \perp$, it implies $\Pi.\mathcal{E}_K(N,A,M) = C$. A nonce-respecting adversary may choose a nonce arbitrarily in encryption, as long as it is different from those already used in previous encryption queries [Rog04]. A nonce in decryption has no restrictions; thus, it may repeat or collide with those used in encryption. For two AE schemes with the same nonce, AD, and message spaces, and the same ciphertext expansion factor, we call them *compatible*. Note that their key spaces may differ. For an AE scheme $\Pi = (\Pi.\mathcal{E}, \Pi.\mathcal{D})$, let \$ denote the random-bit oracle that takes $(N,A,M) \in \mathcal{N}\times\mathcal{AD}\times\mathcal{M}$ and returns random $|\Pi.\mathcal{E}_K(N,A,M)|$ bits. Here, $\perp$ (as an oracle) denotes the decryption oracle that always returns $\perp$. The standard nonce-respecting security notions of Π are defined as follows.

Definition 1 (PRIV and AUTH). *Let $\mathcal{A}_{\mathsf{priv}}$ and $\mathcal{A}_{\mathsf{auth}}$ be nonce-respecting adversaries against Π. PRIV and AUTH advantages are defined as follows.*

$$\mathbf{Adv}^{\mathtt{priv}}_{\Pi}(\mathcal{A}_{\mathsf{priv}}) \coloneqq |\Pr[\mathcal{A}^{\Pi.\mathcal{E}_K}_{\mathsf{priv}} = 1] - \Pr[\mathcal{A}^{\$}_{\mathsf{priv}} = 1]|,$$
$$\mathbf{Adv}^{\mathtt{auth}}_{\Pi}(\mathcal{A}_{\mathsf{auth}}) \coloneqq \Pr[\mathcal{A}^{\Pi.\mathcal{E}_K,\Pi.\mathcal{D}_K}_{\mathsf{auth}} \text{ forges}],$$

The event $[\mathcal{A}^{\Pi.\mathcal{E}_K,\Pi.\mathcal{D}_K}_{\mathsf{auth}}$ forges$]$ means that $\mathcal{A}_{\mathsf{auth}}$ receives some $M' \neq \perp$ from $\Pi.\mathcal{D}_K$. The adversary $\mathcal{A}_{\mathsf{auth}}$ is not allowed to make a trivial forgery, namely a decryption query (N,A,C) after obtaining C by an encryption query (N,A,M).

We say Π is AE-secure if it is secure with respect to both PRIV and AUTH notions.

While nonces used by encryption queries are strictly unique under the PRIV and AUTH notions, it is also important to consider the nonce-misuse setting as introduced by Rogaway and Shrimpton [RS06], namely, encryption queries may repeat nonces, to reflect various practical situations that (erroneously) allow nonce repeat. The security notions capturing this setting are obtained by modifying PRIV and AUTH so that the adversary may repeat nonces in encryption queries but never repeat the entire encryption query (*i.e.*, an encryption query (N,A,M) is always unique). We define these nonce-misusing variations of PRIV and AUTH, written as MR-PRIV and MR-AUTH, as $\mathbf{Adv}^{\mathtt{mr\text{-}priv}}_{\Pi}(\mathcal{A}_{\mathsf{priv}})$ and $\mathbf{Adv}^{\mathtt{mr\text{-}auth}}_{\Pi}(\mathcal{A}_{\mathsf{auth}})$, assuming $\mathcal{A}_{\mathsf{priv}}$ and $\mathcal{A}_{\mathsf{auth}}$ may repeat nonces in encryption. A class of AE schemes that satisfy the MR-PRIV and MR-AUTH notions is called a Misuse-resistant

AE (MRAE). SIV [RS06] is a representative MRAE scheme that combines CMAC-based MAC and a counter mode. We say Π is MRAE-secure if it is secure with respect to both MR-PRIV and MR-AUTH notions.

We write a $(q_e, \sigma_e, \mathtt{tm})$-PRIV adversary to mean a PRIV adversary of q_e encryption queries with σ_e total blocks in encryption queries and $\mathtt{tm}$ time. Similarly, we write a $(q_e, q_d, \sigma_e, \sigma_d, \mathtt{tm})$-AUTH adversary to mean an AUTH-adversary with q_e encryption queries, q_d decryption queries, σ_e total blocks in encryption queries, σ_d total blocks in decryption queries, and $\mathtt{tm}$ time. The same applies to MR-PRIV/MR-AUTH.

Nonce-based encryption. A nonce-based encryption scheme $\Gamma = (\Gamma.\mathcal{E}, \Gamma.\mathcal{D})$, where $\Gamma.\mathcal{E}, \Gamma.\mathcal{D} : \mathcal{K} \times \mathcal{N} \times \mathcal{M} \to \mathcal{M}$, is a plain encryption scheme without authenticity. For correctness, we require $\Gamma.\mathcal{D}_K(N, \Gamma.\mathcal{E}_K(N, M)) = M$ for any (K, N, M). When Γ is tidy, $\Gamma.\mathcal{D}_K(N, C) = M$ implies $\Gamma.\mathcal{E}_K(N, M) = C$. We also define the PRIV notion for Γ and the PRIV advantage $\mathbf{Adv}^{\mathtt{priv}}_{\Gamma}(\mathcal{A}_{\mathsf{priv}})$ analogously to the case of AE.

Committing security. Committing security of Π captures the intractability of finding a valid distinct input tuples, (K, N, A, M) and (K', N', A', M'), that lead to the same ciphertext. Among many notions, we focus on the context-committing security [BH22, CR22], defined as follows.

Definition 2 (Context-committing security). *Let $\Pi = (\Pi.\mathcal{E}, \Pi.\mathcal{D})$ be an AE scheme. The context committing advantage of $\mathcal{A}_{\mathsf{cmt}}$ against Π is defined as*

$$\begin{aligned}\mathbf{Adv}^{\mathtt{cmt}}_{\Pi}(\mathcal{A}_{\mathsf{cmt}}) \coloneqq \Pr[&((K, N, A, M), (K', N', A', M')) \leftarrow \mathcal{A}_{\mathsf{cmt}} : \\ &(K, N, A, M) \neq (K', N', A', M'), \\ &\Pi.\mathcal{E}(K, N, A, M) = \Pi.\mathcal{E}(K', N', A', M')].\end{aligned}$$

Context committing security is the strongest among the committing security notions, as it does not restrict the input tuple beyond distinctness, $(K, N, A, M) \neq (K', N', A', M')$. It is also called CMT-4 [BH22] or full committing [CR22]. We write CMT to mean context committing.

2.2 Additional notations

All the proofs of our proposals are based on the hybrid arguments. For notational convenience, we introduce the following general indistinguishability notions.

Definition 3 (Indistinguishability of AE). *Let $\Pi_i = (\Pi_i.\mathcal{E}, \Pi_i.\mathcal{D})$ for $i = 1, 2$ be compatible AE schemes with a key space $\mathcal{K}_i$.*

$$\begin{aligned}\mathbf{Adv}^{\mathtt{ind}}_{\Pi_1.\mathcal{E}, \Pi_2.\mathcal{E}}(\mathcal{A}) &\coloneqq |\Pr[\mathcal{A}^{\Pi_1.\mathcal{E}_{K_1}} = 1] - \Pr[\mathcal{A}^{\Pi_2.\mathcal{E}_{K_2}} = 1]|, \\ \mathbf{Adv}^{\mathtt{ind}}_{\Pi_1, \Pi_2}(\mathcal{A}') &\coloneqq |\Pr[\mathcal{A}'^{\Pi_1.\mathcal{E}_{K_1}, \Pi_1.\mathcal{D}_{K_1}} = 1] - \Pr[\mathcal{A}'^{\Pi_2.\mathcal{E}_{K_2}, \Pi_2.\mathcal{D}_{K_2}} = 1]|,\end{aligned}$$

where $\mathcal{A}$ and $\mathcal{A}'$ are nonce-respecting, and the probabilities are defined over the adversary's randomness and $K_i \leftarrow\!\!\$\, \mathcal{K}_i$ for $i \in \{1, 2\}$. The adversary $\mathcal{A}'$ is not allowed to make a trivial forgery, namely a decryption query (N, A, C) after obtaining C by an encryption query (N, A, M).

By definition, for any $\mathcal{A}_{\mathsf{priv}}$ and $\mathcal{A}_{\mathsf{auth}}$, we have

$$\mathbf{Adv}^{\mathtt{priv}}_{\Pi}(\mathcal{A}_{\mathsf{priv}}) = \mathbf{Adv}^{\mathtt{ind}}_{\Pi.\mathcal{E},\$}(\mathcal{A}_{\mathsf{priv}}),$$
$$\mathbf{Adv}^{\mathtt{auth}}_{\Pi}(\mathcal{A}_{\mathsf{auth}}) = \mathbf{Adv}^{\mathtt{ind}}_{\Pi,(\Pi.\mathcal{E},\perp)}(\mathcal{A}_{\mathsf{auth}}).$$

For any Π_1, Π_2 and AUTH adversary $\mathcal{A}_{\mathsf{auth}}$,

$$\mathbf{Adv}^{\mathtt{auth}}_{\Pi_1}(\mathcal{A}_{\mathsf{auth}}) - \mathbf{Adv}^{\mathtt{auth}}_{\Pi_2}(\mathcal{A}_{\mathsf{auth}}) \leq \mathbf{Adv}^{\mathtt{ind}}_{\Pi_1,\Pi_2}(\mathcal{A}')$$

holds for some distinguishing adversary $\mathcal{A}'$ using the same number of queries and the total queried blocks as $\mathcal{A}_{\mathsf{auth}}$.

3 Defining Generic Transform

Let $\Omega[*,\mathbb{P}]$ be a generic transform for committing AEs. The first argument takes either an AE scheme or a nonce-based plain encryption scheme, and $\mathbb{P}$ denotes a list of cryptographic primitives specified by the transform Ω. Let $\mathbb{P} = (\mathsf{F}_1,\ldots,\mathsf{F}_a,\mathsf{P}_1,\ldots,\mathsf{P}_b)$ for some positive integers a and b, where F_i denotes a primitive under a standard model (*e.g.*, PRF) and P_j denotes a primitive under an idealized model (*e.g.*, RO). We say $\Omega[*,\mathbb{P}]$ is instantiated if all the primitives in $\mathbb{P}$ are instantiated, *i.e.*, their concrete algorithms are determined. For example, F_1 may be a 128-bit block PRP and an instantiation of F_1 by $\mathsf{AES\text{-}128}$ means that we define $\mathsf{F}_1(x)$ as $\mathsf{AES}(K,x)$ with $K \leftarrow\!\$\, \{0,1\}^{128}$. If P_i is an RO, its instantiation could be typically a popular cryptographic hash function such as $\mathsf{SHA\text{-}2}$ or $\mathsf{SHA\text{-}3}$ [Dwo15].

Our focus is *generic* transforms; thus, we assume Ω can access baseline AE Π only in a black-box way, that is, it can invoke $\Pi.\mathcal{E}$ and $\Pi.\mathcal{D}$ but cannot access any internal functions. If Π uses $\mathbb{P}$ as its primitives, we write $\Pi[\mathbb{P}]$. The goal of a generic transform is to guarantee AE security and committing security after the transform.

Let $\overline{\mathbb{P}}$ denote an instantiation of $\mathbb{P}$. The corresponding instantiation of $\Omega[*,\mathbb{P}]$ is denoted by $\Omega[*,\overline{\mathbb{P}}]$. We consider the following security properties of Ω.

Definition 4. *Let $\Omega[*,\overline{\mathbb{P}}]$ be an instantiation of Ω and let Π be an efficient (i.e., polynomial-time) AE scheme with AE security. The algorithm of Π may depend on $\Omega[*,\overline{\mathbb{P}}]$. We say $\Omega[\Pi,\overline{\mathbb{P}}]$ is broken if any of PRIV/AUTH/CMT security of $\Omega[\Pi,\overline{\mathbb{P}}]$ is lost with $O(1)$ queries and negligible time. We say Ω is uninstantiable if there exists AE-secure ($\overline{\mathbb{P}}$-dependent) Π such that $\Omega[\Pi,\overline{\mathbb{P}}]$ is broken for any $\overline{\mathbb{P}}$. Moreover, we say Ω is conditionally uninstantiable if there exists an AE-secure Π such that $\Omega[\Pi,\overline{\mathbb{P}}]$ is broken for some $\overline{\mathbb{P}}$.*

If Ω is uninstantiable it is also conditionally uninstantiable, but not vice versa.

Remarks. Definition 4 reflects the natural motivation of generic transforms, *i.e.*, a generic transform should maintain security irrespective of the algorithm of a baseline secure AE scheme. If the target transform is uninstantiable or

conditionally uninstantiable, a potentially malicious party could build an AE scheme that is secure as a stand-alone scheme, however, it turns out to be totally broken when transformed into a committing AE. It would be generally non-trivial to tell if a generic transform involving idealized primitives is securely instantiable. Conversely, a provably secure transform that uses only standard-model-secure primitives is inherently immune to the attack against a specific instantiation. That is, there is no gap between the model and the instantiation.

4 Analysis of Generic Transforms

Definition 4 suggests an interesting object to investigate: a generic transform with provable security claims based on idealized-model assumptions on its primitives and standard-model assumptions on AE, which we call the mixed (security) model. A provable security claim based on the mixed model does not generally guarantee the security of its instantiation, and at worst, attacks on instantiations may occur.

This section shows that the worst case occurs with the state-of-the-art generic transforms proposed by CK25 and BBD24. All our counterexamples result in privacy and/or authenticity break, and even plaintext recovery or key recovery, with a few queries. Most of our attacks work against any instantiation of the internal primitives, thus showing they are uninstantiable. The counterexample AE schemes are mostly quite simple and natural; for instance, some use a keyed hash function in the counter mode and build an AE via generic composition. We remark that hash-based encryption schemes have been discussed, *e.g.*, by [HKR01,PRZ99], and have received attention by practitioners, as evidenced by multiple threads on Stack Exchange [STEa, STEb, STEc, STEd].

4.1 Transforms by Chen and Karadžić

At Eurocrypt 2025, Chen and Karadžić proposed a family of generic transforms [CK25] consisting of three basic transforms, HtAE, AEaH, and EtH, and two advanced transforms, AEtH and chaSIV. Their proposals are very clean and simple: most of them invoke a hash function that takes a portion of baseline AE's input/output. The simplicity of their schemes makes them similar to the aforementioned CTX [CR22], but unlike CTX, their schemes are strictly generic. The basic transforms achieve PRIV and AUTH (thus under nonce-respecting setting) and CMT security, and the advanced ones achieve MR-PRIV and MR-AUTH and CMT notions.[5] Some schemes (AEaH, EtH, AEtH) avoid nonce-based rekeying, a common approach to achieving committing security, but this approach is costly. See the paper [CK25] for a more detailed comparison with the previous (generic) transforms.

We show practical PRIV/AUTH attacks against their instantiations, except HtAE. We also present full- and partial-plaintext recovery and key-recovery

[5] They also achieve context discovery security [MLGR23], which is out of our focus.

$\mathsf{AEaH}.\mathcal{E}[\Pi,\mathsf{H}](K,N,A,M)$	$\mathsf{AEaH}.\mathcal{D}[\Pi,\mathsf{H}](K,N,A,(C,T))$
1. $C \leftarrow \Pi.\mathcal{E}(K,N,\varepsilon,M)$ 2. $T \leftarrow \mathsf{H}(K,N,A)$ 3. **return** (C,T)	1. $M \leftarrow \Pi.\mathcal{D}(K,N,\varepsilon,C)$ 2. **if** $M=\perp \vee \mathsf{H}(K,N,A) \neq T$: 3. **return** $\perp$ 4. **return** M
$\mathsf{EtH}.\mathcal{E}[\Gamma,\mathsf{H}](K,N,A,M)$	$\mathsf{EtH}.\mathcal{D}[\Gamma,\mathsf{H}](K,N,A,(C,T))$
1. $C \leftarrow \Gamma.\mathcal{E}(K,N,M)$ 2. $T \leftarrow \mathsf{H}(K,N,A,C)$ 3. **return** (C,T)	1. $T^* \leftarrow \mathsf{H}(K,N,A,C)$ 2. **if** $T^* \neq T$: **return** $\perp$ 3. $M \leftarrow \Gamma.\mathcal{D}(K,N,C)$ 4. **return** M
$\mathsf{HtAE}.\mathcal{E}[\Pi,\mathsf{H}](K,N,A,M)$	$\mathsf{HtAE}.\mathcal{D}[\Pi,\mathsf{H}](K,N,A,(C,T))$
1. $(K_e,T) \leftarrow \mathsf{H}(K,N,A)$ 2. $C \leftarrow \Pi.\mathcal{E}(K_e,N,\varepsilon,M)$ 3. **return** (C,T)	1. $(K_e,T^*) \leftarrow \mathsf{H}(K,N,A)$ 2. **if** $T^* \neq T$: **return** $\perp$ 3. $M \leftarrow \Pi.\mathcal{D}(K_e,N,\varepsilon,C)$ 4. **return** M

Fig. 1: Basic transforms of [CK25]. (Top) AEaH, (Middle) EtH, and (Bottom) HtAE.

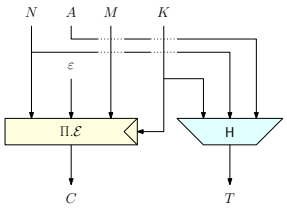


Fig. 2: AEaH.

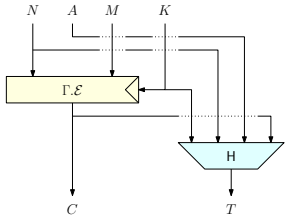


Fig. 3: EtH.

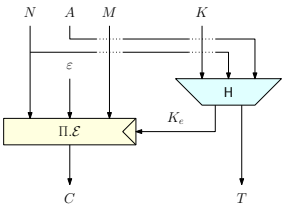


Fig. 4: HtAE.

attacks for some schemes. All our attacks require a few queries. Throughout the section, we assume all the internal idealized primitives have been instantiated. For example, by writing H, we mean a concrete hash function.

4.2 Attacks against AEaH

AE and Hash (AEaH) is a basic transform which uses a hash function $\mathsf{H} : \{0,1\}^* \to \{0,1\}^t$.[6] See Figs. 1 and 2 for the pseudocode and the illustration. Let $\mathsf{AEaH}[\Pi,\mathsf{H}]$ denote AEaH applied to an AE Π. CK25 proved PRIV and AUTH for $\mathsf{AEaH}[\Pi,\mathsf{H}]$ assuming that H is an RO and Π is AE secure. Recall that we allow Π to use H as we consider attacks against instantiations of AEaH (thus H).

Privacy attack. We first describe a privacy attack. Let Π be an Enc-then-MAC AE using H as the primitive. For an mt-bit plaintext $M = (M[1] \| \ldots \| M[m])$,

[6] To be more precise, AEaH assumes H to accept a tuple input in $\mathcal{K} \times \mathcal{N} \times \mathcal{AD}$, hence we assume an injective encoding here. Similar implicit encoding is assumed for other transforms as well.

$\mathsf{AEtH}.\mathcal{E}[\Pi,\mathsf{H}](K,N,A,M)$	$\mathsf{AEtH}.\mathcal{D}[\Pi,\mathsf{H}](K,N,A,(C,T))$
1. $C \leftarrow \Pi.\mathcal{E}(K,N,A,M)$ 2. $T \leftarrow \mathsf{H}(K,N,A,\mathtt{lsb}_\tau(C))$ 3. **return** (C,T)	1. $T^* \leftarrow \mathsf{H}(K,N,A,\mathtt{lsb}_\tau(C))$ 2. **if** $T^* \neq T$: **return** $\perp$ 3. $M \leftarrow \Pi.\mathcal{D}(K,N,A,C)$ 4. **return** M
$\mathsf{chaSIV}.\mathcal{E}[\Gamma,\mathsf{F},\mathsf{H}](K,N,A,M)$	$\mathsf{chaSIV}.\mathcal{D}[\Gamma,\mathsf{F},\mathsf{H}](K,N,A,(C,T))$
1. $K_e \leftarrow \mathsf{F}(K,N)$ 2. $T \leftarrow \mathsf{H}(K,N,A,M)$ 3. $N_e \leftarrow \mathtt{lsb}_r(T)$ 4. $C \leftarrow \Gamma.\mathcal{E}(K_e,N_e,M)$ 5. **return** (C,T)	1. $K_e \leftarrow \mathsf{F}(K,N)$ 2. $N_e \leftarrow \mathtt{lsb}_r(T)$ 3. $M \leftarrow \Gamma.\mathcal{D}(K_e,N_e,C)$ 4. **if** $M = \perp$: **return** $\perp$ 5. $T^* \leftarrow \mathsf{H}(K,N,A,M)$ 6. **if** $T^* \neq T$: **return** $\perp$ 7. **return** M

Fig. 5: Advanced transforms of [CK25]. (Top) AEtH and (Bottom) chaSIV.

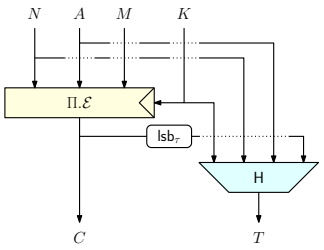


Fig. 6: AEtH.

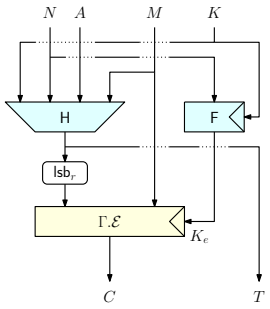


Fig. 7: chaSIV.

$|M[i]| = t$, the tuple (N,A,M) is encrypted as $\Pi.\mathcal{E}(K,N,A,M) \to C = (\widetilde{C},\widetilde{T})$, $\widetilde{C} = (\widetilde{C}[1] \,\|\, \ldots \,\|\, \widetilde{C}[m])$ such that

$$\widetilde{C}[i] = \mathsf{H}(K,N,\langle i \rangle) \oplus M[i] \text{ for } i \in [1..m].$$

$\widetilde{T}$ is the tag of Π defined as $\widetilde{T} = \mathsf{MAC}_K(N,A,\widetilde{C})$ for some MAC function MAC. MAC may be arbitrarily instantiated by H or any other additional primitive, using the key K, possibly with a domain separation from the encryption part. A secure AE can be easily realized assuming H_K is a PRF by provable security analysis of Enc-then-MAC [NRS14]. The PRF assumption on H_K would be reasonable if H is a reasonable instantiation of RO. The privacy attack works as follows.

1. Encrypt (N,A,M) with any N, $A = \langle 1 \rangle$, $|M| = mt$ for some $m \geq 1$. Receive $(C = (\widetilde{C},\widetilde{T}),T) = \mathsf{AEaH}[\Pi,\mathsf{H}].\mathcal{E}(K,N,A,M)$.
2. Output 1 (decide as real) if $\widetilde{C}[1] \oplus T = M[1]$ holds. Otherwise, output 0.

Clearly, because $T = \mathsf{H}(K,N,A) = \mathsf{H}(K,N,\langle 1 \rangle)$, $\widetilde{C}[1] \oplus T = M[1]$ holds with probability 1 in the real world, while with probability $\sim 1/2^t$ in the ideal world. The adversary's advantage is almost 1. Moreover, a (partial) plaintext recovery attack is possible; observing (C,T) for the aforementioned $(N,A = \langle 1 \rangle,M)$ with

unknown M will reveal $M[1]$. Generally, if A is chosen among $\{\langle j\rangle\}_{j\in[1..m]}$, any single plaintext block could be recovered. When $m = 1$, it means full plaintext recovery.

Authenticity attack. Let Γ be a nonce-based encryption scheme. We instantiate Π as a variant of GCM combining Γ and GHASH and H. The encryption algorithm of Π is as follows. Suppose $\mathcal{N} = \{0,1\}^n \setminus \{\mathtt{cst}_1, \mathtt{cst}_2\}$ for some distinct constants $\mathtt{cst}_1$ and $\mathtt{cst}_2$. Let $\Pi.\mathcal{E}(K, N, A, M) = C = (\widetilde{C}, \widetilde{T})$, where

$$L_i = \mathsf{H}(K, \mathtt{cst}_i, \varepsilon) \text{ for } i = 1, 2, \quad \widetilde{C} = \Gamma.\mathcal{E}(L_1, N, M),$$
$$\widetilde{T} = \mathsf{GHASH}(L_2, (A, \widetilde{C})) \oplus \mathsf{H}(K, N, A).$$

The above Π is provably AE-secure if Γ is PRIV secure and H_K is a PRF; the entire scheme can be seen as a variant of GCM with two keys, and the keys L_1, L_2 and the last masking value $\mathsf{H}(K, N, A)$ will look independent and random thanks to the domain separation. Its provable security bounds will be easily derived using the known results on generic composition and GCM [NRS14,IOM12]. Recall that the baseline Π fixes AD to empty when composed. The transformed scheme $\mathsf{AEaH}[\Pi, \mathsf{H}]$ is vulnerable to an authenticity attack. The procedure is similar to the classical nonce-reuse attack against GCM [Jou06] and enables almost universal forgery (arbitrary manipulation of the ciphertext). Assume AD $A = \varepsilon$.

1. Encrypt $(N, A(= \varepsilon), M)$ and receive $(C, T) = \mathsf{AEaH}[\Pi, \mathsf{H}].\mathcal{E}(K, N, \varepsilon, M)$, where $C = (\widetilde{C}, \widetilde{T}) = \Pi.\mathcal{E}(K, N, \varepsilon, M)$ and $T = \mathsf{H}(K, N, \varepsilon)$.
2. Since $\widetilde{T} \oplus T = \Big(\mathsf{GHASH}(L_2, (\varepsilon, \widetilde{C})) \oplus \mathsf{H}(K, N, \varepsilon)\Big) \oplus \mathsf{H}(K, N, \varepsilon)$ is equal to the GHASH evaluation, $\mathsf{GHASH}(L_2, (\varepsilon, \widetilde{C}))$, the adversary recovers L_2.
3. Decrypt $(N, A(= \varepsilon), C', T)$ such that $C' = (\widetilde{C}', \widetilde{T}')$, $\widetilde{C}' \neq \widetilde{C}$, and $\widetilde{T}' = \mathsf{GHASH}(L_2, (\varepsilon, \widetilde{C}')) \oplus \mathsf{H}(K, N, \varepsilon)$.

The decryption query will be accepted with probability 1.

4.3 Attacks against EtH

Enc then Hash (EtH) is a basic transform which uses a hash function $\mathsf{H} : \{0,1\}^* \to \{0,1\}^t$. The baseline scheme Γ may be a nonce-based encryption or an AE. See Figs. 1 and 3 for the pseudocode and the illustration. Let $\mathsf{EtH}[\Gamma, \mathsf{H}]$ denote EtH applied to Γ. CK25 proved PRIV and AUTH for $\mathsf{EtH}[\Gamma, \mathsf{H}]$ assuming that H is an RO and Γ is PRIV secure.

Privacy attack. Our privacy attack utilizes a H-based CTR mode with a certain trick in the argument. Let $\mathcal{N} = \{0,1\}^*$, $\mathcal{M} = \{0,1\}^{\leq m \cdot t}$ for some $m > 0$, and $\mathcal{AD} = \{0,1\}^*$. We define $\Gamma.\mathcal{E} : \mathcal{K} \times \mathcal{N} \times \mathcal{M} \to \mathcal{M}$ as a variant of CTR. Let $\mathtt{cst} \in \{0,1\}^*$ be a constant. Let $c = \log_2 m$.

For $(N, M) \in \mathcal{N} \times \mathcal{M}$ such that $(M[1], \ldots, M[m]) \xleftarrow{t} M$, $C = \Gamma.\mathcal{E}(K, N, M)$ where $C = C[1] \,\|\, \ldots \,\|\, C[m]$ is defined as follows.

$$C[i] = \mathtt{msb}_{|M[i]|}(\mathsf{H}(K, \mathtt{cst}, N, \langle i-1\rangle_c)) \oplus M[i].$$

This Γ is PRIV secure if H_K is a PRF; however, the placement of nonce in the argument of H is inconsistent with that of H call inside EtH. This inconsistency is exploited as follows.

1. Encrypt $(N = \mathtt{cst}, A, M)$ for arbitrary $A \neq \mathtt{cst}$ and M such that $|M| = \ell$ for some $\ell \in [1..c]$ (possibly 1). Receive (C, T) where $|C| = \ell$ and $T = \mathsf{H}(K, \mathtt{cst}, A, C)$.
2. Let $j \in [1..2^c]$ so that $\langle j-1 \rangle_c = C$. Encrypt $(N' = A, A', M')$ for arbitrary A' and M' such that $|M'| \geq j \cdot t$. Receive (C', T').
3. Check if $C'[j] \oplus M'[j]$ equals to T. If it holds, decide it as real, and otherwise decide it as ideal.

Since

$$C'[j] = \mathsf{H}(K, \mathtt{cst}, N', \langle j-1 \rangle_c) \oplus M'[j] = \mathsf{H}(K, \mathtt{cst}, A, C) \oplus M'[j] = T \oplus M'[j]$$

holds, the check will always succeed in the real world, whereas it succeeds with a negligible probability in the ideal world. Thus, the advantage of the attack is almost 1.

Authenticity attack. Similar to the case of AEaH, an authenticity attack is possible when Γ is a counter mode using H: we first parse M into t-bit blocks, $M[1], M[2], \ldots$, and encrypt as $C[i] = \mathsf{H}(K, N, \mathtt{cst}, \langle i \rangle_c) \oplus M[i]$ for some constant $\mathtt{cst} \in \{0,1\}^*$ and some sufficiently large $c > 0$. Then $\mathsf{H}(K, N, \mathtt{cst}, \langle i \rangle_c)$ for $i = 1, 2, \ldots$, can be learned by an encryption query, resulting in the following attack.

1. Encrypt (N, A, M) for any (N, A) and $|M| \geq m \cdot t$ for some positive integer m. Receive $(C, T) = \mathsf{EtH}[\Pi, \mathsf{H}].\mathcal{E}(K, N, A, M)$.
2. Pick any $i \in [1..m]$. Decrypt $(N, A' = \mathtt{cst}, C' = \langle i \rangle_c, T' = M[i] \oplus C[i])$.

The decryption query will be accepted as $M[i] \oplus C[i] = \mathsf{H}(K, N, \mathtt{cst}, \langle i \rangle_c) = \mathsf{H}(K, N, A', C')$.

4.4 Attack against AEtH

AE then Hash (AEtH) is an advanced transform which uses a hash function $\mathsf{H} : \{0,1\}^* \to \{0,1\}^t$. See Figs. 5 and 6 for the pseudocode and the illustration. Let $\mathsf{AEtH}[\Pi, \mathsf{H}]$ denote AEtH applied to an AE Π. CK25 proved MR-PRIV and MR-AUTH for $\mathsf{AEtH}[\Pi, \mathsf{H}]$ assuming that H is an RO and Π is MRAE secure. AEtH is called MRAE-preserving for this security guarantee.

Let Π be an SIV-like MRAE scheme using some variable-input-length PRF $\mathsf{F} : \mathcal{K} \times \{0,1\}^* \to \{0,1\}^\tau$ and H. For encryption of (N, A, M) for $|M| = mt$, $\Pi.\mathcal{E}$ first computes $V = \mathsf{F}(K, N, A, M)$ (assuming implicit encoding). After t-bit parsing of M, each block is encrypted as $C[i] = M[i] \oplus \mathsf{H}(K, N, A, V \oplus \langle i-1 \rangle_\tau)$ for $i \in [1..m]$. The output is $C = (C[1] \,\|\, \ldots \,\|\, C[m] \,\|\, V)$. The provable MRAE security of this Π is established if MAC part (F_K) and Encryption part (H_K)

are computationally independent [RS06]. In practice, this could be ensured by using a different algorithm, *e.g.*, using SHA-3 for H and CMAC-AES for F_K, or instantiating F by H with an adequate domain separation.

When the above Π is transformed by AEtH, an encryption query (N, A, M) generates $(C, T) = \mathsf{AEtH}[\Pi, \mathsf{H}].\mathcal{E}(K, N, A, M)$, where $C = (C[1] \,\|\, \ldots \,\|\, C[m] \,\|\, V)$ and $T = \mathsf{H}(K, N, A, \mathtt{lsb}_\tau(C))$ with $\mathtt{lsb}_\tau(C) = V = \mathsf{F}(K, N, A, M)$.

Privacy attack. The privacy attack is similar to the previous cases: since $C[1] \oplus M[1] = \mathsf{H}(K, N, A, V \oplus \langle 0 \rangle_\tau (= 0^\tau)) = T$ holds, the adversary can distinguish (C, T) from random with high probability. In the context of a plaintext recovery attack, the adversary can recover $M[1]$.

We have not found any authenticity attack.

4.5 Attack against chaSIV

Committing hash-based SIV (chaSIV) is an advanced transform which uses a hash function $\mathsf{H} : \{0,1\}^* \to \{0,1\}^t$ and a PRF $\mathsf{F} : \mathcal{K} \times \mathcal{N} \to \mathcal{K}_e$. The baseline scheme Γ is either a nonce-based encryption or an AE, taking key space $\mathcal{K}_e$ and nonce space $\mathcal{N}_e = \{0,1\}^r$. Following CK25, we assume $\mathcal{K} = \mathcal{K}_e = \{0,1\}^\kappa$, which ensures that the key size does not change before and after the transform. See Figs. 5 and 7 for the pseudocode and the illustration. Let $\mathsf{chaSIV}[\Gamma, \mathsf{F}, \mathsf{H}]$ denote chaSIV applied to Γ. CK25 proved MR-PRIV and MR-AUTH for $\mathsf{chaSIV}[\Gamma, \mathsf{F}, \mathsf{H}]$ assuming that H is an RO, F is a PRF, and Γ is PRIV secure, thus chaSIV is *MRAE-lifting*.

Unlike other transforms, the primitives of the chaSIV transform already rely on the mixed model. We consider the case that F uses the instantiation of H as a primitive.

Privacy attack. Let $\kappa \le t$ and $r \le t$. Let $\mathsf{F}(K, N)$ be $\mathtt{msb}_\kappa(\mathsf{H}(K, N, \mathtt{cst}_1, \mathtt{cst}_2))$ for some constants $\mathtt{cst}_1$ and $\mathtt{cst}_2$ (they may depend on N). Following other cases, the PRF assumption on F would be reasonable. To break privacy, the adversary encrypts (N, A, M) for $A = \mathtt{cst}_1$ and $M = \mathtt{cst}_2$. Then the output $(C, T) = \mathsf{chaSIV}.\mathcal{E}[\Gamma, \mathsf{H}, \mathsf{F}](K, N, A, M)$ reveals K_e as $K_e = \mathtt{msb}_\kappa(T)$. This attack implies a complete privacy break. Moreover, if the adversary does not know M, it implies full plaintext recovery. The attack works regardless of the algorithm used by Γ.

The above PRIV attack is devastating (full plaintext recovery for any Γ). On the other hand, we could not find an authenticity attack. Moreover, our PRIV attack needs specific conditions on the instantiation (*i.e.*, F uses H). The latter implies that chaSIV is conditionally uninstantiable. We have not found an authenticity attack.

4.6 PACT and comPACT transforms

Bhattacharjee *et al.* [BBD24] proposed PACT and comPACT. Their structures are similar to CTX. Both transforms use a hash function $\mathsf{H} : \{0,1\}^* \to \{0,1\}^t$

$\mathsf{PACT}.\mathcal{E}[\Pi, \mathsf{E}, \mathsf{H}](K, N, A, M)$	$\mathsf{PACT}.\mathcal{D}[\Pi, \mathsf{E}, \mathsf{H}](K, N, A, (\widetilde{C}, T))$
1. $(\widetilde{C}, \widetilde{T}) \leftarrow \Pi.\mathcal{E}(K, N, \boxed{A}, M)$	1. $K^* \leftarrow \mathsf{H}(K, N, A)$
2. $K^* \leftarrow \mathsf{H}(K, N, A)$	2. $\widetilde{T} \leftarrow \mathsf{E}^{-1}(K^*, T)$
3. $T \leftarrow \mathsf{E}(K^*, \widetilde{T})$	3. $M \leftarrow \Pi.\mathcal{D}(K, N, \boxed{A}, (\widetilde{C}, \widetilde{T}))$
4. **return** $(\widetilde{C}, T)$	4. **return** M

Fig. 8: PACT and comPACT transforms. comPACT is obtained by replacing the gray-boxed AD, $\boxed{A}$, with an empty string ε.

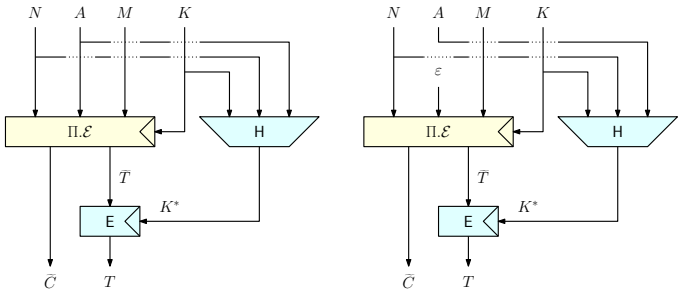


Fig. 9: (Left) PACT and (Right) comPACT.

and a block cipher $\mathsf{E} : \mathcal{K}_b \times \mathcal{M}_b \to \mathcal{M}_b$, $\mathcal{K}_b = \{0,1\}^t$ and $\mathcal{M}_b = \{0,1\}^\tau$. We require $\tau = t$. The τ-bit tag part of the baseline AE is encrypted by E_{K^*}, where $K^* = \mathsf{H}(K, N, A)$. On decryption, the tag for the baseline AE is first recovered by $\mathsf{E}^{-1}_{K^*}$. Unlike CTX, this mechanism enables the use of the baseline AE in a black-box way. Fig. 9 illustrates PACT and comPACT. Fig. 8 shows their pseudocodes. comPACT is identical to PACT except comPACT fixes the AD input of Π to ε. The technique is similar to CTY [BH24] and improves efficiency as AD is only processed once by H. Let $\mathsf{PACT}[\Pi, \mathsf{E}, \mathsf{H}]$ and $\mathsf{comPACT}[\Pi, \mathsf{E}, \mathsf{H}]$ denote the PACT and comPACT transforms applied to Π. Bhattacharjee *et al.* proved (MR-)PRIV and (MR-)AUTH of $\mathsf{PACT}[\Pi, \mathsf{E}, \mathsf{H}]$ assuming collision resistance of H, E being an ideal cipher, and (MR-)PRIV and (MR-)AUTH of Π. Similarly, they proved PRIV and AUTH of $\mathsf{comPACT}[\Pi, \mathsf{E}, \mathsf{H}]$ under the same assumptions on E and H as for PACT, assuming Π is AE-secure. They proved both PACT and comPACT are CMT secure with the same assumption as above, with an additional mild assumption on Π, called Ciphertext Collision (CC) [BBD24].

As in the case of chaSIV, PACT and comPACT employ the mixed model for their primitives. We show attacks against any instantiation of them.

Privacy attack against PACT. Let Π be a variant of SIV. For encrypting a tuple (N, A, M), Π first performs the MAC computation to determine the tag $\widetilde{T}$ as a (MAC) function of (K, N, A, M). Next, it derives the internal key as $K^* = \mathsf{H}(K, N, A)$. After parsing M into τ-bit blocks, the i-th ciphertext block is obtained as $\widetilde{C}[i] = M[i] \oplus \mathsf{E}(K^*, \widetilde{T} \oplus \langle i-1 \rangle_\tau)$. As this Π follows the template of SIV, it is provably secure by adequately instantiating the MAC function

and E. Note that E's instantiation must be reasonable for an ideal cipher. We observe that $\widetilde{C}[1] = M[1] \oplus \mathsf{E}(K^*, \widetilde{T} \oplus \langle 0 \rangle_\tau (= \widetilde{T})) = M[1] \oplus T$ holds for any valid encryption transcript $(N, A, M, (\widetilde{C}, T))$ of PACT for $|M| \geq \tau$. This observation implies a privacy break and plaintext recovery of $M[1]$ if M is unknown. Full plaintext recovery occurs if $|M| \leq \tau$.

Authenticity attack against PACT. Consider a variant of GCM using Universal hash-then-Encrypt. Let Π be an AE scheme based on a nonce-based encryption scheme $\Gamma : \mathcal{K} \times \mathcal{N}' \times \mathcal{M} \to \mathcal{M}$ for $\mathcal{N}' = \mathcal{N} \times \mathcal{AD}$, a block cipher $\mathsf{E} : \mathcal{K}_b \times \mathcal{M}_b \to \mathcal{M}_b$, and GHASH defined over $\mathrm{GF}(2^\tau)$. Let N' denote the tuple (N, A) and let $\Pi.\mathcal{E}(K, N, A, M) = (\widetilde{C}, \widetilde{T})$, where

$$\widetilde{C} = \Gamma(K, N', M), \quad \widetilde{T} = \mathsf{E}^{-1}(K^*, V), \quad V = \mathsf{GHASH}(L, (A, \widetilde{C})),$$

and $L = \mathsf{KDF}(K)$ for some KDF $\mathsf{KDF} : \mathcal{K} \to \{0,1\}^\tau$, $K^* = \mathsf{H}(K, N, A)$ as in the case of privacy attack.[7] The black-box security of Π is easy to prove by adequately instantiating KDF, Γ and E. If we want Π to be a nonce-based AE, Γ can be a CTR mode with nonce N'. If we want Π to be an MRAE, so that it is a meaningful counterexample of PACT, Γ can be realized by a tweakable wide-block cipher $\widetilde{E} : \mathcal{K} \times \mathcal{N}' \times \mathcal{M} \to \mathcal{M}$, such as $\mathsf{HCTR2}$ [WFW05, CHB21], with tweak N'; $\Gamma(K, N', M) = \widetilde{E}(K, N', M) = \widetilde{C}$. Such Π will ensure MR-PRIV as any distinct input (N, A, M) will produce a random $\widetilde{C}$. This Π will also ensure MR-AUTH for its Enc-then-MAC structure. After transformation, $\mathsf{PACT}[\Pi, \mathsf{E}, \mathsf{H}]$ computes T as $\mathsf{E}(K^*, \widetilde{T}) = \mathsf{E}(K^*, \mathsf{E}^{-1}(K^*, V)) = V$ thus it reveals V, the output of GHASH. The knowledge of V allows us to derive the GHASH key L. The attack is as follows.

1. Encrypt (N, A, M) and receive $(\widetilde{C}, T) = \mathsf{PACT}[\Pi, \mathsf{E}, \mathsf{H}].\mathcal{E}(K, N, A, M)$.
2. From $T = V = \mathsf{GHASH}(L, (A, \widetilde{C}))$, recover L.
3. Choose $\widetilde{C}' \neq \widetilde{C}$ such that $\mathsf{GHASH}(L, (A, \widetilde{C})) = \mathsf{GHASH}(L, (A, \widetilde{C}'))$. Decrypt $(N, A, (\widetilde{C}', T))$ which will be accepted with probability 1.

Finding $\widetilde{C}'$ in Step 3 is straightforward, see *e.g.* [Jou06, FIM20].

Attacking $\mathsf{comPACT}$. The above privacy and authenticity attacks also work for $\mathsf{comPACT}$, with a minor modification to the authenticity attack.

4.7 Key Recovery Attacks

Our attacks shown above exploit AE schemes that appear natural due to their use of strong primitives in a generic composition. We show that if we adopt slightly artificial forms of (authenticated) encryption, some transforms allow

[7] While we try to mimic GCM in its use of GHASH, feeding A to GHASH may be redundant as K^* already contains information on A. Variations of the attack are possible when GHASH takes only $\widetilde{C}$ or takes $\widetilde{C}$ and a constant instead of A.

even total key recovery. The baseline encryption schemes we used are provably secure on their own; hence, they are theoretically valid counterexamples. The trick is simple; we use a counter mode with an additional (and unnecessary) Key Blinding (KB) mechanism. See below for details.

Key recovery against AEaH. The most straightforward key-recovery attack is against AEaH. The privacy attack against AEaH uses CTR mode $\widetilde{C}[i] \leftarrow M[i] \oplus \mathsf{H}(K, N, \langle i \rangle)$. For key recovery, we instead use CTR with key blinding, dubbed CTRKB, such that $\widetilde{C}[i] \leftarrow M[i] \oplus \mathsf{H}(K, N, \langle i \rangle) \oplus K$. The privacy attack is converted into a key recovery attack; K is recovered by taking the sum $\widetilde{C}[1] \oplus M[1] \oplus T$. It is reasonable to assume that CTRKB maintains standard model security if the inner H is a well-analyzed cryptographic hash function such as SHA-2/3, which is also a reasonable candidate for RO instantiation.[8]

Key recovery against EtH. We modify the PRIV attack so that it uses a variant of CTRKB such that $C[i] = \mathtt{msb}_{|M[i]|}(\mathsf{H}(K, \mathtt{cst}, N, \langle i-1 \rangle_c)) \oplus K \oplus M[i]$. The attack is obtained by changing the last step of the PRIV attack, as in the case of AEaH.

Key recovery against AEtH. We modify the PRIV attack so that it uses a variant of CTRKB such that $C[i] = M[i] \oplus \mathsf{H}(K, N, A, V \oplus \langle i-1 \rangle) \oplus K$.

Key recovery against PACT/comPACT. We modify the PRIV attack so that it uses a variant of CTRKB such that $\widetilde{C}[i] = M[i] \oplus \mathsf{E}_{K^*}(T \oplus \langle i-1 \rangle) \oplus K$.

Key recovery against chaSIV. Our key recovery attack against chaSIV is more involved than others. The simple key blinding technique seems inapplicable, since the F output is secret and the H algorithm should not have an exploitable structure, as originally assumed to be a monolithic RO. Let π be a κ-bit cryptographic permutation for even κ. Let $r = \kappa$. We assume $\mathcal{M} = \{0,1\}^{\kappa/2}$ for simplicity. Recall $|K| = |K_e| = \kappa$. Let $\mathtt{cst}_1 \in \mathcal{AD}$ and $\mathtt{cst}_2 \in \mathcal{M}$ be some constant bit strings used by the PRIV attack. Let $\mathsf{F} : \mathcal{K} \times \mathcal{N} \to \mathcal{K}$ and $\Gamma : \mathcal{K} \times \mathcal{N} \times \mathcal{M} \to \mathcal{M}$ be defined as follows. First, we parse the key as $K = (K[0] \,\|\, K[1])$ where $|K[0]| = |K[1]| = \kappa/2$. Γ is a variant of CTR mode using π with truncation. The encryption is

$$\Gamma.\mathcal{E}(K_e, N_e, M) = C = \mathtt{msb}_{\kappa/2}(\pi(N_e \oplus K_e)) \oplus M.$$

The decryption of Γ is analogously defined. We next design F so that it behaves badly when combined with the above Γ. In more detail, we define

$$\mathsf{F}(K, N) = K_e = \mathtt{msb}_\kappa(V_N) \oplus S[\mathtt{msb}_1(N)],$$

where $S[i] = \pi^{-1}(K[i] \,\|\, 0^{\kappa/2})$ for $i \in \{0,1\}$ and $V_N = \mathsf{H}(K, N, \mathtt{cst}_1, \mathtt{cst}_2)$.

[8] In a general context, it may happen that additional key blinding collapses original security. A typical example is CTR built on an Even-Mansour cipher.

We show this combination indeed leaks the key. For an encryption query $(N, A = \mathtt{cst}_1, M = \mathtt{cst}_2)$ with $\mathtt{msb}_1(N) = 0$, the adversary receives (C, T) and recovers $K[0]$ since

$$\begin{aligned} C \oplus M &= \mathtt{msb}_{\kappa/2}(\pi(N_e \oplus K_e)) \\ &= \mathtt{msb}_{\kappa/2}(\pi(\underbrace{\mathtt{msb}_\kappa(V_N)}_{N_e} \oplus \underbrace{S[0] \oplus \mathtt{msb}_\kappa(V_N)}_{K_e})) \\ &= \mathtt{msb}_{\kappa/2}(\pi(\pi^{-1}(K[0] \,\|\, 0^{\kappa/2}))) \\ &= \mathtt{msb}_{\kappa/2}(K[0] \,\|\, 0^{\kappa/2}) \\ &= K[0]. \end{aligned}$$

Another encryption query with N' such that $\mathtt{msb}_1(N') = 1$ will reveal $K[1]$. Thus, using two encryption queries with different MSBs, the full K will be recovered.

These "contrived" functions F and Γ themselves would be secure if π and H are reasonable instantiations of a public random permutation and an RO. Let us give a brief analysis. Our F takes the sum of $S[\mathtt{msb}_1(N)]$ and V_N, where the latter is deemed as an output of a PRF $\mathsf{H}(K, *, \mathtt{cst}_1, \mathtt{cst}_2)$. Assuming π and H are algorithmically different, adding $S[\mathtt{msb}_1(N)]$ to V_N should cause no problem on PRF security. A provable security bound can be easily obtained if we make idealized assumptions about them. Our Γ is effectively a CTR mode of truncated permutation π with input masking by the key. If π is a public random permutation, Γ has $(\kappa/2 - \log_2 \kappa)$-bit security [HNS16]. We remark that the design of this counterexample is subtle: we cannot use Even-Mansour for Γ because in that case K_e will remain in the keystream $M \oplus C$, hence it will not be PRIV secure. If we skip truncation for Γ, it will be insecure, as an inverse permutation query will reveal K_e. Moreover, if we use the entire K instead of half to generate $S[i]$, it reveals only part of K, and the rest remains unrevealed.

This example of Γ should look somewhat unnatural. There would be more straightforward, intuitive examples. However, our example works irrespective of the instantiation of H. Thus, we think it is a meaningful attack against an instantiation of chaSIV.

5 Provably Secure Generic Transforms

5.1 Fixing CK25 transforms

Our attacks exploit the gap between the mixed model and its instantiation. A natural next step is to address the gap; however, this is inherently difficult if we keep the mixed model.

As mentioned in Sect. 3, any transform solely relying on the standard model is inherently safe. Following this principle, we present how to modify the broken CK25 transforms to obtain standard model proofs.[9] Note that uninstantiable

[9] We do not consider PACT and comPACT, as they are not strictly generic due to their requirement of an additional property for AE and potential revisions mentioned by the designers (private communication).

transforms (AEaH, EtH, AEtH, PACT, and comPACT) cannot have standard model proofs, hence modifications are necessary for them. To do so, we keep changes minimal and, in particular, consider whether the original scheme requires nonce-based rekeying, since this can be a key factor in determining overall performance, as observed by CK25. Concretely, we present fixed versions for AEaH, AEtH, EtH, and chaSIV. For HtAE, we show that standard-model security is achievable without changing the algorithm. As mentioned in the Introduction, two of them (AEaH-KS and AEtH-KS) are the first transforms that are

1. Strictly generic,
2. Context committing,
3. Free from nonce-based rekeying, and
4. Entirely standard-model secure

known in the literature (see also Tables 1 and 2). In other words, we salvage AEaH and AEtH, those equipped with Properties 1 to 3 but uninstantiable, and add Property 4, keeping Properties 1 to 3.

Random IVs do not help. Our attacks exploit the fact that the broken schemes allow the baseline AE to fully simulate the computation performed by the transform. To thwart these attacks, it looks plausible to generate a random IV, *IV*, inside the transform and give *IV only* to the hash function. However, this generally does not work, since a random IV is part of the encryption output. The adversary can learn the random IV used by an encryption query and provide it to AE (*e.g.*, as part of AD) via another query. Some broken examples could be shown, for instance, taking AEaH.

5.2 Fixing AEaH

We present how to fix AEaH. For this, we assume H has an explicit key input $\mathcal{K}$ (for simplicity: the key space can be different from that of AE), making $\mathsf{H} : \mathcal{K} \times \{0,1\}^* \to \{0,1\}^t$. We introduce a PRG $\mathsf{G} : \mathcal{K} \to \mathcal{K} \times \mathcal{K}$. The fix is consistent and simple: we apply $\mathsf{G}(K) = (K_1, K_2)$ and use K_1 for Π and K_2 for H. We call this technique *Key separation* (KS). Since KS is needed only once, it is preferable to nonce-based rekeying. The fixed scheme, AEaH-KS, is presented by Fig. 10 (top). We prove that AEaH-KS is secure assuming G is an RCR-PRG and H is a CR-PRF.

Theorem 1. *Let* $\mathcal{A}_{\mathsf{priv}}$ *and* $\mathcal{A}_{\mathsf{auth}}$ *be* $(q_e, \sigma_e, \mathtt{tm})$*-PRIV and* $(q_e, q_d, \sigma_e, \sigma_d, \mathtt{tm})$*-AUTH adversaries against* AEaH-KS$[\Pi, \mathsf{G}, \mathsf{H}]$*. Let* $q = q_e + q_d$ *and* $\sigma = \sigma_e + \sigma_d$*. Then*

$$\mathbf{Adv}^{\mathtt{priv}}_{\mathsf{AEaH\text{-}KS}[\Pi,\mathsf{G},\mathsf{H}]}(\mathcal{A}_{\mathsf{priv}}) \le \mathbf{Adv}^{\mathtt{prg}}_{\mathsf{G}}(\mathcal{A}_{\mathsf{G}}) + \mathbf{Adv}^{\mathtt{prf}}_{\mathsf{H}}(\mathcal{A}_{\mathsf{H}}) + \mathbf{Adv}^{\mathtt{priv}}_{\Pi}(\mathcal{A}_{\Pi})$$
$$\mathbf{Adv}^{\mathtt{auth}}_{\mathsf{AEaH\text{-}KS}[\Pi,\mathsf{G},\mathsf{H}]}(\mathcal{A}_{\mathsf{auth}}) \le \mathbf{Adv}^{\mathtt{prg}}_{\mathsf{G}}(\mathcal{A}'_{\mathsf{G}}) + \mathbf{Adv}^{\mathtt{prf}}_{\mathsf{H}}(\mathcal{A}'_{\mathsf{H}}) + \mathbf{Adv}^{\mathtt{auth}}_{\Pi}(\mathcal{A}'_{\Pi}) + \frac{q_d}{2^t},$$

$\mathsf{AEaH\text{-}KS}.\mathcal{E}[\Pi,\mathsf{G},\mathsf{H}](K,N,A,M)$	$\mathsf{AEaH\text{-}KS}.\mathcal{D}[\Pi,\mathsf{G},\mathsf{H}](K,N,A,(C,T))$
1. $(K_1,K_2) \leftarrow \mathsf{G}(K)$ 2. $C \leftarrow \Pi.\mathcal{E}(K_1,N,\varepsilon,M)$ 3. $T \leftarrow \mathsf{H}(K_2,N,A)$ 4. **return** (C,T)	1. $(K_1,K_2) \leftarrow \mathsf{G}(K)$ 2. $M \leftarrow \Pi.\mathcal{D}(K_1,N,\varepsilon,C)$ 3. **if** $M = \perp \vee \mathsf{H}(K_2,N,A) \neq T$: 4. **return** $\perp$ 5. **return** M
$\mathsf{AEtH\text{-}KS}.\mathcal{E}[\Pi,\mathsf{G},\mathsf{H}](K,N,A,M)$	$\mathsf{AEtH\text{-}KS}.\mathcal{D}[\Pi,\mathsf{G},\mathsf{H}](K,N,A,(C,T))$
1. $(K_1,K_2) \leftarrow \mathsf{G}(K)$ 2. $C \leftarrow \Pi.\mathcal{E}(K_1,N,A,M)$ 3. $T \leftarrow \mathsf{H}(K_2,N,A,\mathtt{lsb}_\tau(C))$ 4. **return** (C,T)	1. $(K_1,K_2) \leftarrow \mathsf{G}(K)$ 2. $T^* \leftarrow \mathsf{H}(K_2,N,A,\mathtt{lsb}_\tau(C))$ 3. **if** $T^* \neq T$: **return** $\perp$ 4. $M \leftarrow \Pi.\mathcal{D}(K_1,N,A,C)$ 5. **return** M
$\mathsf{EtH\text{-}KS}.\mathcal{E}[\Gamma,\mathsf{G},\mathsf{H}](K,N,A,M)$	$\mathsf{EtH\text{-}KS}.\mathcal{D}[\Gamma,\mathsf{G},\mathsf{H}](K,N,A,(C,T))$
1. $(K_1,K_2) \leftarrow \mathsf{G}(K)$ 2. $C \leftarrow \Gamma.\mathcal{E}(K_1,N,M)$ 3. $T \leftarrow \mathsf{H}(K_2,N,A,C)$ 4. **return** (C,T)	1. $(K_1,K_2) \leftarrow \mathsf{G}(K)$ 2. $T^* \leftarrow \mathsf{H}(K_2,N,A,C)$ 3. **if** $T^* \neq T$: **return** $\perp$ 4. $M \leftarrow \Gamma.\mathcal{D}(K_1,N,C)$ 5. **return** M

Fig. 10: Fixed versions of CK25 transforms by means of key separation. (Top) AEaH-KS, (Middle) AEtH-KS, and (Bottom) EtH-KS.

holds, where $\mathcal{A}_\mathsf{G}$ *and* $\mathcal{A}'_\mathsf{G}$ *are* $\mathtt{tm}'$*-PRG adversaries.* $\mathcal{A}_\mathsf{H}$ *is a* $(q_e,\sigma_e,\mathtt{tm}')$*-PRF adversary,* $\mathcal{A}_\Pi$ *is a* $(q_e,\sigma_e,\mathtt{tm}')$*-PRIV adversary,* $\mathcal{A}'_\mathsf{H}$ *is a* $(q,\sigma,\mathtt{tm}')$*-PRF adversary, and* $\mathcal{A}'_\Pi$ *is a* $(q_e,q_d,\sigma_e,\sigma_d,\mathtt{tm}')$*-AUTH adversary.*

Theorem 2. *Let* $\mathcal{A}_\mathsf{cmt}$ *be* $\mathtt{tm}$*-CMT adversary against* $\mathsf{AEaH\text{-}KS}[\Pi,\mathsf{G},\mathsf{H}]$.

$$\mathbf{Adv}^{\mathtt{cmt}}_{\mathsf{AEaH\text{-}KS}[\Pi,\mathsf{G},\mathsf{H}]}(\mathcal{A}_\mathsf{cmt}) \leq \mathbf{Adv}^{\mathtt{cr}}_{\mathsf{H}}(\mathcal{A}_\mathsf{H}) + \mathbf{Adv}^{\mathtt{rcr}}_{\mathsf{G}}(\mathcal{A}_\mathsf{G}),$$

where $\mathcal{A}_\mathsf{H}$ *is a* $\mathtt{tm}'$*-CR adversary and* $\mathcal{A}_\mathsf{G}$ *is a* $\mathtt{tm}'$*-RCR adversary.*

5.3 Proofs of Theorems 1 and 2 (AEaH-KS)

Proof of Theorem 1. Let $\$_\mathsf{G}$ be a uniform random source over $\mathcal{K}\times\mathcal{K}$ and let $\mathsf{R}_\mathsf{H} : \{0,1\}^* \to \{0,1\}^t$ be a variable-input-length URF. Let H_K denote the oracle that returns $\mathsf{H}(K,N,A)$ with uniform key $K \leftarrow\!\!\$\, \mathcal{K}$ on input $(N,A) \in \mathcal{N}\times\mathcal{AD}$. Let Θ_1, Θ_2, and Θ_3 denote $\mathsf{AEaH\text{-}KS}[\Pi,\mathsf{G},\mathsf{H}]$, $\mathsf{AEaH\text{-}KS}[\Pi,\$_\mathsf{G},\mathsf{H}]$, and $\mathsf{AEaH\text{-}KS}[\Pi,\$_\mathsf{G},\mathsf{R}_\mathsf{H}]$ respectively. We have

$$\begin{aligned}
&\mathbf{Adv}^{\mathtt{priv}}_{\mathsf{AEaH\text{-}KS}[\Pi,\mathsf{G},\mathsf{H}]}(\mathcal{A}_\mathsf{priv})\\
&\leq \mathbf{Adv}^{\mathtt{ind}}_{\Theta_1.\mathcal{E},\Theta_2.\mathcal{E}}(\mathcal{A}_\mathsf{priv}) + \mathbf{Adv}^{\mathtt{ind}}_{\Theta_2.\mathcal{E},\Theta_3.\mathcal{E}}(\mathcal{A}_\mathsf{priv}) + \mathbf{Adv}^{\mathtt{ind}}_{\Theta_3.\mathcal{E},\$}(\mathcal{A}_\mathsf{priv})\\
&\leq \mathbf{Adv}^{\mathtt{ind}}_{\mathsf{G},\$_\mathsf{G}}(\mathcal{A}_\mathsf{G}) + \mathbf{Adv}^{\mathtt{ind}}_{\mathsf{H}_K,\mathsf{R}_\mathsf{H}}(\mathcal{A}_\mathsf{H}) + \mathbf{Adv}^{\mathtt{ind}}_{\Theta_3.\mathcal{E},\$}(\mathcal{A}_\mathsf{priv})\\
&\leq \underbrace{\mathbf{Adv}^{\mathtt{prg}}_{\mathsf{G}}(\mathcal{A}_\mathsf{G})}_{\text{by definition}} + \underbrace{\mathbf{Adv}^{\mathtt{prf}}_{\mathsf{H}}(\mathcal{A}_\mathsf{H})}_{\text{by definition}} + \mathbf{Adv}^{\mathtt{priv}}_{\Pi}(\mathcal{A}'_\Pi).
\end{aligned}$$

Here, $\mathbf{Adv}^{\mathtt{ind}}_{\Theta_3,\$}(\mathcal{A}_{\Theta_3}) \leq \mathbf{Adv}^{\mathtt{ind}}_{\Pi.\mathcal{E},\$}(\mathcal{A}'_\Pi)$ holds as the T part in any encryption response is always independently uniform and random for both games (because

$T = \mathsf{R}_{\mathsf{H}}(N, A)$ in Θ_3 and nonce does not repeat), thus not contributing to the advantage. This proves the privacy claim.

For authenticity bound, we have

$$\begin{aligned}
&\mathbf{Adv}^{\mathtt{auth}}_{\mathsf{AEaH\text{-}KS}[\Pi,\mathsf{G},\mathsf{H}]}(\mathcal{A}_{\mathsf{auth}}) \\
&\leq \mathbf{Adv}^{\mathtt{ind}}_{\Theta_1,\Theta_2}(\mathcal{A}'_{\mathsf{auth}}) + \mathbf{Adv}^{\mathtt{auth}}_{\Theta_2}(\mathcal{A}_{\mathsf{auth}}) \\
&\leq \mathbf{Adv}^{\mathtt{ind}}_{\Theta_1,\Theta_2}(\mathcal{A}'_{\mathsf{auth}}) + \mathbf{Adv}^{\mathtt{ind}}_{\Theta_2,\Theta_3}(\mathcal{A}''_{\mathsf{auth}}) + \mathbf{Adv}^{\mathtt{auth}}_{\Theta_3}(\mathcal{A}_{\mathsf{auth}}) \\
&\leq \mathbf{Adv}^{\mathtt{ind}}_{\mathsf{G},\$_{\mathsf{G}}}(\mathcal{A}_{\mathsf{G}}) + \mathbf{Adv}^{\mathtt{ind}}_{\mathsf{H}_K,\mathsf{R}_{\mathsf{H}}}(\mathcal{A}_{\mathsf{H}}) + \mathbf{Adv}^{\mathtt{auth}}_{\Theta_3}(\mathcal{A}_{\mathsf{auth}}) \\
&= \mathbf{Adv}^{\mathtt{prg}}_{\mathsf{G}}(\mathcal{A}_{\mathsf{G}}) + \mathbf{Adv}^{\mathtt{prf}}_{\mathsf{H}}(\mathcal{A}_{\mathsf{H}}) + \mathbf{Adv}^{\mathtt{auth}}_{\Theta_3}(\mathcal{A}_{\mathsf{auth}}) \qquad (1)
\end{aligned}$$

The last term $\mathbf{Adv}^{\mathtt{auth}}_{\Theta_3}(\mathcal{A}_{\mathsf{auth}}) \coloneqq \mathbf{Adv}^{\mathtt{ind}}_{\Theta_3,(\Theta_3.\mathcal{E},\perp)}(\mathcal{A}_{\mathsf{auth}})$ needs further hybrids. For clarity, we introduce the following attributes of decryption queries. A decryption query (N', A', C', T') is said to be Π-trivial if there exists an encryption query-response tuple (N, A, M, C, T) such that $(N' = N, A' \neq A, C' = C)$, and H-trivial if there exists an encryption query-response tuple (N, A, M, C, T) such that $(N' = N, A' = A, C' \neq C)$. We remark that a trivial decryption query is prohibited in its entirety: thus, if $(N' = N, A' = A, C' = C)$ holds, it means $T' \neq T$. Such a query will never be successful (*i.e.*, the response is always $\perp$). Therefore, we can exclude such decryption queries. A decryption query which is not Π-trivial is called Π-nontrivial, including *e.g.*, the case that encryption queries do not use N'. Clearly, the sets of Π-trivial queries and H-trivial queries are exclusive, and the latter is a subset of Π-nontrivial queries.

For any keyed function (or oracle) $F_K : \mathcal{X} \to \mathcal{Y}$, let $\mathsf{vrf}[F_K] : \mathcal{X} \times \mathcal{Y} \to \{\top, \perp\}$ be the verification oracle defined as $\mathsf{vrf}[F_K](X, Y) = \top$ if $F_K(X) = Y$ and $\mathsf{vrf}[F_K](X, Y) = \perp$ if $F_K(X) \neq Y$.

We decompose the decryption routine into two sub-oracles, O_1^D and O_2^D, where O_1^D performs AE decryption and O_2^D performs verification regarding H, namely $\mathsf{vrf}[\mathsf{H}_K]$. The response to a decryption query is determined by calling the sub-oracles, and is a tuple rather than a single variable (either the decrypted message or $\perp$). Let $\Pi.\mathcal{D}'_K$ denote a decryption oracle with AD fixed to empty, *i.e.*, $\Pi.\mathcal{D}'_K(N, C) = \Pi.\mathcal{D}_K(N, \varepsilon, C)$. The original decryption routine receiving a query (N, A, C, T) is reinterpreted as follows, assuming $O_1^D = \Pi.\mathcal{D}'_{K_1}$ and $O_2^D = \mathsf{vrf}[\mathsf{H}_{K_2}]$.

1. $O_2^D(N, A, T) \to Y_2$
2. If $Y_2 = \perp$, return $\perp$
3. $O_1^D(N, C) \to Y_1$
4. Return Y_1

Observe that the adversary needs to break both O_1^D and O_2^D. Keeping this feature, we modify the routine as follows.

1. If (N, A, C, T) is Π-trivial, $O_2^D(N, A, T) \to Y_2$ and return (ε, Y_2)
2. Else $O_1^D(N, C) \to Y_1$
3. If $Y_1 \neq \perp$, $O_2^D(N, A, T) \to Y_2$ and return (Y_1, Y_2)
4. Else return $(\perp, \perp)$

It is straightforward to see that this modification does not reduce the chance of win: the output is a tuple (Y_1, Y_2) in the modified routine, and the original response is always recovered from the output tuple, and as in the original routine, the adversary is successful only if it breaks both O_1^D and O_2^D, including the trivial win against O_i^D. Note that a decryption query leading to a trivial win against both sub-oracles is impossible by definition. With this modification, any Π-nontrivial query will be first processed by O_1^D. We write (O_1^D, O_2^D) to mean the decryption oracle. Let $\Theta_3^1 = (\Theta_3.\mathcal{E}, \Theta_3^1.\mathcal{D})$, where $\Theta_3^1.\mathcal{D} = (\Pi.\mathcal{D}', \mathsf{vrf}[\mathsf{R}_\mathsf{H}])$. Similarly, let $\Theta_3^2 = (\Theta_3.\mathcal{E}, \Theta_3^2.\mathcal{D})$, where $\Theta_3^2.\mathcal{D} = (\perp, \mathsf{vrf}[\mathsf{R}_\mathsf{H}])$ and let $\Theta_3^3 = (\Theta_3.\mathcal{E}, \Theta_3^3.\mathcal{D})$, where $\Theta_3^3.\mathcal{D} = (\perp, \perp)$. Clearly, Θ_3^1 and Θ_3^3 (in the modified routine) are the counterparts of Θ_3 and $(\Theta_3.\mathcal{E}, \perp)$ (in the original routine). Since the modified routine augments the original decryption oracle response,

$$\mathbf{Adv}^{\mathtt{ind}}_{\Theta_3,(\Theta_3.\mathcal{E},\perp)}(\mathcal{A}_{\mathsf{auth}}) \le \mathbf{Adv}^{\mathtt{ind}}_{\Theta_3^1,\Theta_3^2}(\mathcal{A}_{\mathsf{auth}}) + \mathbf{Adv}^{\mathtt{ind}}_{\Theta_3^2,\Theta_3^3}(\mathcal{A}_{\mathsf{auth}}) \tag{2}$$

holds. For bounding $\mathbf{Adv}^{\mathtt{ind}}_{\Theta_3^1,\Theta_3^2}(\mathcal{A}_{\mathsf{auth}})$, we observe that distinguishing is possible only by a Π-nontrivial query, since a Π-trivial query is processed by R_H for both worlds. Moreover, once the adversary receives $Y_1 \neq \perp$, this deterministically indicates that the world is Θ_3^1. Once the adversary receives $Y_1 = \perp$ the response is identical, $(\perp, \perp)$. That is, Y_2 does not contribute to the advantage. Thus,

$$\mathbf{Adv}^{\mathtt{ind}}_{\Theta_3^1,\Theta_3^2}(\mathcal{A}_{\mathsf{auth}}) \le \mathbf{Adv}^{\mathtt{auth}}_{\Pi}(\mathcal{A}'_{\Pi}). \tag{3}$$

For bounding $\mathbf{Adv}^{\mathtt{ind}}_{\Theta_3^2,\Theta_3^3}(\mathcal{A}_{\mathsf{auth}})$, we observe that a Π-nontrivial query will not help as it results in $(\perp, \perp)$ response for both worlds. Thus, Π-trivial queries, which do not contain H-trivial queries, only help. This implies that the game reduces to a MAC unforgeability game involving R_H with (N, A) as the message. Since R_H is completely random, the forgery probability is $q_d/2^t$: thus,

$$\mathbf{Adv}^{\mathtt{ind}}_{\Theta_3^2,\Theta_3^3}(\mathcal{A}_{\mathsf{auth}}) \le \frac{q_d}{2^t}. \tag{4}$$

Combining Eqs. (1), (2), (3), and (4), we conclude the proof of Theorem 1.

Proof of Theorem 2. Suppose $Z = ((K, N, A, M), (K', N', A', M'), (C, T))$ be the final output of $\mathcal{A}_{\mathsf{cmt}}$ with the pair of corresponding ciphertext and tag (C, T). Let $(K_1, K_2) = \mathsf{G}(K)$ and $(K'_1, K'_2) = \mathsf{G}(K')$. Assuming Z is valid, we consider two cases. First, if $K = K'$, we have $(K_1, K_2) = (K'_1, K'_2)$. When $(N, A) = (N', A')$ holds, it implies $M = M'$, hence Z is not valid. For Z to be valid, $\mathsf{H}(K_2, N, A) = \mathsf{H}(K'_2, N', A')(= T)$ must hold for some $(N, A) \neq (N', A')$, implying a collision of H outputs. Second, suppose $K \neq K'$. The subcase $[K_2 = K'_2]$ reduces to a right-half collision on G. The subcase $[K_2 \neq K'_2]$ implies a collision on H. Overall, for any case, the adversary needs to find either a collision on H or a right-half collision on G, or more concretely,

$$\begin{aligned}\mathbf{Adv}^{\mathtt{cmt}}_{\mathsf{AEaH\text{-}KS}[\Pi,\mathsf{G},\mathsf{H}]}(\mathcal{A}_{\mathsf{cmt}}) &\le \max\{\mathbf{Adv}^{\mathtt{cr}}_{\mathsf{H}}(\mathcal{A}_{\mathsf{H}}), \mathbf{Adv}^{\mathtt{rcr}}_{\mathsf{G}}(\mathcal{A}_{\mathsf{G}})\},\\ &\le \mathbf{Adv}^{\mathtt{cr}}_{\mathsf{H}}(\mathcal{A}_{\mathsf{H}}) + \mathbf{Adv}^{\mathtt{rcr}}_{\mathsf{G}}(\mathcal{A}_{\mathsf{G}}),\end{aligned}$$

which concludes the proof.

5.4 Fixing AEtH

We can fix AEtH in the same manner as AEaH. We apply KS to AEtH and obtain AEtH-KS as shown in Fig. 10 (middle). We prove its security assuming G is an RCR-PRG and H is a CR-PRF.

Theorem 3. *Let $\mathcal{A}_{\mathsf{priv}}$ and $\mathcal{A}_{\mathsf{auth}}$ be $(q_e, \sigma_e, \mathtt{tm})$-MR-PRIV and $(q_e, q_d, \sigma_e, \sigma_d, \mathtt{tm})$-MR-AUTH adversaries against $\mathsf{AEtH\text{-}KS}[\Pi, \mathsf{G}, \mathsf{H}]$. Let $q = q_e + q_d$ and $\sigma = \sigma_e + \sigma_d$. We have*

$$\mathbf{Adv}^{\mathtt{mr\text{-}priv}}_{\mathsf{AEtH\text{-}KS}[\Pi,\mathsf{G},\mathsf{H}]}(\mathcal{A}_{\mathsf{priv}}) \le \mathbf{Adv}^{\mathtt{prg}}_{\mathsf{G}}(\mathcal{A}_{\mathsf{G}}) + \mathbf{Adv}^{\mathtt{prf}}_{\mathsf{H}}(\mathcal{A}_{\mathsf{H}}) + \mathbf{Adv}^{\mathtt{mr\text{-}priv}}_{\Pi}(\mathcal{A}_{\Pi}) + \frac{q_e^2}{2^{\tau+1}},$$

$$\mathbf{Adv}^{\mathtt{mr\text{-}auth}}_{\mathsf{AEtH\text{-}KS}[\Pi,\mathsf{G},\mathsf{H}]}(\mathcal{A}_{\mathsf{auth}}) \le \mathbf{Adv}^{\mathtt{prg}}_{\mathsf{G}}(\mathcal{A}'_{\mathsf{G}}) + \mathbf{Adv}^{\mathtt{mr\text{-}auth}}_{\Pi}(\mathcal{A}'_{\Pi}),$$

where $\mathcal{A}_{\mathsf{G}}$ and $\mathcal{A}'_{\mathsf{G}}$ are $\mathtt{tm}'$-PRG adversaries, $\mathcal{A}_{\mathsf{H}}$ is a $(q_e, \sigma_e, \mathtt{tm}')$-PRF adversary, $\mathcal{A}_{\Pi}$ is a $(q_e, \sigma_e, \mathtt{tm}')$-MR-PRIV adversary, and $\mathcal{A}'_{\Pi}$ is a $(q_e, q_d, \sigma_e, \sigma_d, \mathtt{tm}')$-MR-AUTH adversary.

Theorem 4. *Let $\mathcal{A}_{\mathsf{cmt}}$ be a $\mathtt{tm}$-CMT adversary against $\mathsf{AEtH\text{-}KS}[\Pi, \mathsf{G}, \mathsf{H}]$.*

$$\mathbf{Adv}^{\mathtt{cmt}}_{\mathsf{AEtH\text{-}KS}[\Pi,\mathsf{G},\mathsf{H}]}(\mathcal{A}_{\mathsf{cmt}}) \le \mathbf{Adv}^{\mathtt{cr}}_{\mathsf{H}}(\mathcal{A}_{\mathsf{H}}) + \mathbf{Adv}^{\mathtt{rcr}}_{\mathsf{G}}(\mathcal{A}_{\mathsf{G}}),$$

where $\mathcal{A}_{\mathsf{H}}$ is a $\mathtt{tm}'$-CR adversary and $\mathcal{A}_{\mathsf{G}}$ is a $\mathtt{tm}'$-RCR adversary.

5.5 Proofs of Theorems 3 and 4 (AEtH-KS)

Proof of Theorem 3 As in Theorem 1, the hybrid arguments that replace G with a random source $\$_{\mathsf{G}}$ and H with a URF R_{H} are trivial. We derive the PRIV bound for Θ_1, which denotes $\mathsf{AEtH}[\Pi, \$_{\mathsf{G}}, \mathsf{R}_{\mathsf{H}}]$. Let $\Theta_2 = \mathsf{AEtH}[\$_{\Pi}, \$_{\mathsf{G}}, \mathsf{R}_{\mathsf{H}}]$ denote a slight variant of Θ_1 whose encryption generates C in a completely random way irrespective of the input. For MR-PRIV, we have

$$\begin{aligned}\mathbf{Adv}^{\mathtt{mr\text{-}priv}}_{\Theta_1}(\mathcal{A}_{\mathsf{priv}}) &\le \mathbf{Adv}^{\mathtt{ind}}_{\Theta_1.\mathcal{E},\Theta_2.\mathcal{E}}(\mathcal{A}_{\mathsf{priv}}) + \mathbf{Adv}^{\mathtt{ind}}_{\Theta_2.\mathcal{E},\$}(\mathcal{A}_{\mathsf{priv}}) \\ &\le \mathbf{Adv}^{\mathtt{mr\text{-}priv}}_{\Pi}(\mathcal{A}_{\Pi}) + \frac{q_e^2}{2^{\tau+1}}, \qquad (5)\end{aligned}$$

where the first term of Eq. (5) holds since $T \leftarrow \mathsf{R}_{\mathsf{H}}(K_2, N, A, \mathtt{lsb}_{\tau}(C))$ for $K_2 \leftarrow\!\!\$\, \mathcal{K}$ holds for both worlds (thus the advantage will not increase if T is omitted from the response). The second term holds since, as long as the $\mathtt{lsb}_{\tau}(C)$ part does not collide, the outputs of both $\Theta_2.\mathcal{E}$ and \$ are equivalent, and C is random for both worlds. By adding computational terms with respect to G and H, this proves the MR-PRIV bound.

For the MR-AUTH bound, we define Θ_1 as $\mathsf{AEtH}[\Pi, \$_{\mathsf{G}}, \mathsf{H}]$. We also define Θ_2 as $\mathsf{AEtH}[(\Pi.\mathcal{E}, \perp), \$_{\mathsf{G}}, \mathsf{H}]$. Note that $\Theta_2.\mathcal{D}$ is equivalent to $\perp$ oracle as it always returns $\perp$ irrespective of the computation of H during decryption, because Π is modified so that it always returns $\perp$ when decrypting. To distinguish Θ_1 from Θ_2, the adversary needs to invoke $M \neq \perp$ at the last line of the decryption routine

(Fig. 5), implying the authenticity break of Π. In addition, a decryption query implying a trivial win against Π, *i.e.*, (N, A, C, T) such that $\Pi.\mathcal{E}(K, N, A, M) \to C$ is already computed in an encryption query, will not win by definition. Thus we have

$$\mathbf{Adv}^{\mathtt{mr\text{-}auth}}_{\Theta_1}(\mathcal{A}_{\mathsf{auth}}) \leq \mathbf{Adv}^{\mathtt{ind}}_{\Theta_1,\Theta_2}(\mathcal{A}'_{\mathsf{auth}}) \leq \mathbf{Adv}^{\mathtt{mr\text{-}auth}}_{\Pi}(\mathcal{A}'_{\Pi}).$$

Adding PRG and PRF terms to the above bound concludes the proof.

Proof of Theorem 4. Following the proof of Theorem 2, suppose $Z = ((K, N, A, M), (K', N', A', M'), (C, T))$ be a valid input tuple determined by the adversary, and the corresponding ciphertext-tag tuple (C, T). Let $(K_1, K_2) = \mathsf{G}(K)$ and $(K'_1, K'_2) = \mathsf{G}(K')$. When $K = K'$, $(N, A) = (N', A')$ implies $M = M'$, thus Z is not valid (attack never succeeds). When $K = K'$ and $(N, A) \neq (N', A')$ hold, it implies H's collision. When $K \neq K'$, if $K_2 \neq K'_2$ occurs we need a collision of H, and if $K_2 = K'_2$ occurs it implies a right-half collision of G. Thus, a valid Z either requires a collision on H or a right-half collision of G. This concludes the proof.

5.6 Fixing EtH

As for EtH, we apply the key separation. The resulting scheme, denoted by EtH-KS, is almost identical to the generic composition, namely Enc-then-MAC. Our analysis is almost identical to the existing security analysis of Enc-then-MAC: for black-box security by Namprempre *et al.* [NRS14], and for committing security by Struck and Weishäupl [SW24] and Bhaumik *et al.* [BCC+24].

Theorem 5. *Let $\mathcal{A}_{\mathsf{priv}}$ and $\mathcal{A}_{\mathsf{auth}}$ be $(q_e, \sigma_e, \mathtt{tm})$-PRIV and $(q_e, q_d, \sigma_e, \sigma_d, \mathtt{tm})$-AUTH adversaries against $\mathsf{EtH\text{-}KS}[\Gamma, \mathsf{G}, \mathsf{H}]$. Let $q = q_e + q_d$ and $\sigma = \sigma_e + \sigma_d$. We have*

$$\begin{aligned}
\mathbf{Adv}^{\mathtt{priv}}_{\mathsf{EtH\text{-}KS}[\Gamma,\mathsf{G},\mathsf{H}]}(\mathcal{A}_{\mathsf{priv}}) &\leq \mathbf{Adv}^{\mathtt{prg}}_{\mathsf{G}}(\mathcal{A}_{\mathsf{G}}) + \mathbf{Adv}^{\mathtt{prf}}_{\mathsf{H}}(\mathcal{A}_{\mathsf{H}}) + \mathbf{Adv}^{\mathtt{priv}}_{\Gamma}(\mathcal{A}_{\Gamma}), \\
\mathbf{Adv}^{\mathtt{auth}}_{\mathsf{EtH\text{-}KS}[\Gamma,\mathsf{G},\mathsf{H}]}(\mathcal{A}_{\mathsf{auth}}) &\leq \mathbf{Adv}^{\mathtt{prg}}_{\mathsf{G}}(\mathcal{A}'_{\mathsf{G}}) + \mathbf{Adv}^{\mathtt{prf}}_{\mathsf{H}}(\mathcal{A}'_{\mathsf{H}}) + \frac{q_d}{2^t},
\end{aligned}$$

where $\mathcal{A}_{\mathsf{G}}$ and $\mathcal{A}'_{\mathsf{G}}$ are $\mathtt{tm}'$-PRG adversaries, $\mathcal{A}_{\mathsf{H}}$ is a $(q_e, \sigma_e, \mathtt{tm}')$-PRF adversary, $\mathcal{A}'_{\mathsf{H}}$ is a $(q, \sigma, \mathtt{tm}')$-PRF adversary, and $\mathcal{A}_{\Gamma}$ is a $(q_e, \sigma_e, \mathtt{tm}')$-PRIV adversary.

Theorem 6. *Let $\mathcal{A}_{\mathsf{cmt}}$ be a $\mathtt{tm}$-CMT adversary against $\mathsf{EtH\text{-}KS}[\Gamma, \mathsf{G}, \mathsf{H}]$. Then,*

$$\mathbf{Adv}^{\mathtt{cmt}}_{\mathsf{EtH\text{-}KS}[\Gamma,\mathsf{G},\mathsf{H}]}(\mathcal{A}_{\mathsf{cmt}}) \leq \mathbf{Adv}^{\mathtt{cr}}_{\mathsf{H}}(\mathcal{A}_{\mathsf{H}}) + \mathbf{Adv}^{\mathtt{rcr}}_{\mathsf{G}}(\mathcal{A}_{\mathsf{G}})$$

holds, where $\mathcal{A}_{\mathsf{H}}$ is a $\mathtt{tm}'$-CR adversary and $\mathcal{A}_{\mathsf{G}}$ is a $\mathtt{tm}'$-RCR adversary.

Appendix A shows the proofs of Theorems 5 and 6.

chaSIV-NKS.$\mathcal{E}[\Gamma,\mathsf{F},\mathsf{H}](K,N,A,M)$	chaSIV-NKS.$\mathcal{D}[\Gamma,\mathsf{F},\mathsf{H}](K,N,A,(C,T))$
1. $K_e \leftarrow \mathsf{F}(K, 0 \,\|\, N)$	1. $K_e \leftarrow \mathsf{F}(K, 0 \,\|\, N)$
2. $K_h \leftarrow \mathsf{F}(K, 1 \,\|\, N)$	2. $K_h \leftarrow \mathsf{F}(K, 1 \,\|\, N)$
3. $T \leftarrow \mathsf{H}(K_h, A, M)$	3. $N_e \leftarrow \mathtt{lsb}_r(T)$
4. $N_e \leftarrow \mathtt{lsb}_r(T)$	4. $M \leftarrow \Gamma.\mathcal{D}(K_e, N_e, C)$
5. $C \leftarrow \Gamma.\mathcal{E}(K_e, N_e, M)$	5. **if** $M = \bot$: **return** $\bot$
6. **return** (C, T)	6. $T^* \leftarrow \mathsf{H}(K_h, A, M)$
	7. **if** $T^* \neq T$: **return** $\bot$
	8. **return** M

Fig. 11: A fixed version of chaSIV, chaSIV-NKS.

5.7 Fixing chaSIV

Several options to fix chaSIV could be considered. Adding KS on top is probably the most intuitive option, though we propose a simpler, more efficient alternative. This reuses the original components and roughly preserves the original complexity; the nonce is processed by F and H in the original, while the fixed version processes it twice by F. Our fixed version, chaSIV-NKS (for Nonce-based Key Separation), is shown in Fig. 11.

Theorem 7. *Let $\mathcal{A}_{\mathsf{priv}}$ and $\mathcal{A}_{\mathsf{auth}}$ be $(q_e, \sigma_e, \mathtt{tm})$-MR-PRIV and $(q_e, q_d, \sigma_e, \sigma_d, \mathtt{tm})$-MR-AUTH adversaries against* chaSIV-NKS$[\Gamma,\mathsf{F},\mathsf{H}]$. *Let $q = q_e + q_d$ and $\sigma = \sigma_e + \sigma_d$. Throughout the MR-PRIV game, let Q_e denote the number of distinct nonces among the encryption queries. We assume that i-th nonce is repeated R_i times for $i \in [1..Q_e]$. Let $R_e^* := \max_{i\in[1..Q_e]} R_i$. For the MR-AUTH game, we assume Γ is tidy. Let Q_{ed} denote the number of distinct nonces among the union of encryption and decryption queries. We define R_i for $i \in [1..Q_{ed}]$ as the number of queries using i-th nonce and define $R_{ed}^* := \max_{i\in[1..Q_{ed}]} R_i$.*

$$\begin{aligned}
\mathbf{Adv}^{\mathtt{mr\text{-}priv}}_{\text{chaSIV-NKS}[\Gamma,\mathsf{F},\mathsf{H}]}(\mathcal{A}_{\mathsf{priv}}) &\le \mathbf{Adv}^{\mathtt{prf}}_{\mathsf{F}}(\mathcal{A}_{\mathsf{F}}) + Q_e \cdot \mathbf{Adv}^{\mathtt{prf}}_{\mathsf{H}}(\mathcal{A}_{\mathsf{H}}) \\
&\quad + Q_e \cdot \mathbf{Adv}^{\mathtt{priv}}_{\Gamma}(\mathcal{A}_{\Gamma}) + \frac{\sum_{i\in[1..Q_e]} R_i^2}{2^r}, \\
\mathbf{Adv}^{\mathtt{mr\text{-}auth}}_{\text{chaSIV-NKS}[\Gamma,\mathsf{F},\mathsf{H}]}(\mathcal{A}_{\mathsf{auth}}) &\le \mathbf{Adv}^{\mathtt{prf}}_{\mathsf{F}}(\mathcal{A}'_{\mathsf{F}}) + Q_{ed} \cdot \mathbf{Adv}^{\mathtt{prf}}_{\mathsf{H}}(\mathcal{A}'_{\mathsf{H}}) + \frac{q_d}{2^t},
\end{aligned}$$

where $\mathcal{A}_{\mathsf{F}}$ is a $(2Q_e, \mathtt{tm}')$-PRF adversary, $\mathcal{A}'_{\mathsf{F}}$ is a $(2Q_{ed}, \mathtt{tm}')$-PRF adversary, $\mathcal{A}_{\mathsf{H}}$ is a $(R_e^*, \sigma_e, \mathtt{tm}')$-PRF adversary, $\mathcal{A}'_{\mathsf{H}}$ is a $(R_{ed}^*, \sigma, \mathtt{tm}')$-PRF adversary, and $\mathcal{A}_{\Gamma}$ is a $(R_e^*, \sigma_e, \mathtt{tm}')$-PRIV adversary.

Theorem 8. *Let $\mathcal{A}_{\mathsf{cmt}}$ be a $\mathtt{tm}$-CMT adversary against* chaSIV-NKS$[\Gamma,\mathsf{F},\mathsf{H}]$. *Then*

$$\mathbf{Adv}^{\mathtt{cmt}}_{\text{chaSIV-NKS}[\Gamma,\mathsf{F},\mathsf{H}]}(\mathcal{A}_{\mathsf{cmt}}) \le \mathbf{Adv}^{\mathtt{cr}}_{\mathsf{H}}(\mathcal{A}_{\mathsf{H}}) + \mathbf{Adv}^{\mathtt{rcr}}_{\mathsf{F}}(\mathcal{A}_{\mathsf{F}})$$

holds, where $\mathcal{A}_{\mathsf{H}}$ is a $\mathtt{tm}'$-CR adversary and $\mathcal{A}_{\mathsf{F}}$ is a $\mathtt{tm}'$-RCR adversary.

Appendix B shows the proofs of Theorems 7 and 8.

5.8 HtAE under the standard model

HtAE uses a hash function $\mathsf{H} : \{0,1\}^* \to \mathcal{K} \times \{0,1\}^t$. For encryption input (N, A, M), $\mathsf{HtAE}.\mathcal{E}_K[\Pi, \mathsf{H}]$ first computes $(K_e, T) \leftarrow \mathsf{H}(K, N, A)$ and $C \leftarrow \Pi.\mathcal{E}_{K_e}(N, \varepsilon, M)$, and outputs (C, T). The decryption is analogously defined, see Fig. 1. As described, we did not find attacks against the original HtAE, even though it still relies on the mixed model, *i.e.*, H is treated as an RO. We prove the standard model security of HtAE without changing the algorithm, but changing the assumption on H. In more detail, we assume $\mathsf{H} : \mathcal{K} \times \mathcal{N} \times \mathcal{A} \to \mathcal{K} \times \{0,1\}^t$ is a PRF with key $K \leftarrow\!\!\$\, \mathcal{K}$ and is right-half collision resistant. For clarity, we call this transform $\mathsf{HtAE\text{-}ST}$.

Theorem 9. *Let $\mathcal{A}_{\mathsf{priv}}$ and $\mathcal{A}_{\mathsf{auth}}$ be $(q_e, \sigma_e, \mathtt{tm})$-PRIV and $(q_e, q_d, \sigma_e, \sigma_d, \mathtt{tm})$-AUTH adversaries against $\mathsf{HtAE\text{-}ST}[\Pi, \mathsf{H}]$. Let Q_d denote the number of distinct nonces in decryption queries. Let $q = q_e + q_d$ and $\sigma = \sigma_e + \sigma_d$. Then*

$$\mathbf{Adv}^{\mathtt{priv}}_{\mathsf{HtAE\text{-}ST}[\Pi,\mathsf{H}]}(\mathcal{A}_{\mathsf{priv}}) \le \mathbf{Adv}^{\mathtt{prf}}_{\mathsf{H}}(\mathcal{A}_{\mathsf{H}}) + q_e \mathbf{Adv}^{\mathtt{priv}}_{\Pi}(\mathcal{A}_{\Pi}),$$
$$\mathbf{Adv}^{\mathtt{auth}}_{\mathsf{HtAE\text{-}ST}[\Pi,\mathsf{H}]}(\mathcal{A}_{\mathsf{auth}}) \le \mathbf{Adv}^{\mathtt{prf}}_{\mathsf{H}}(\mathcal{A}'_{\mathsf{H}}) + Q_d \mathbf{Adv}^{\mathtt{auth}}_{\Pi}(\mathcal{A}'_{\Pi}) + \frac{q_d}{2^t}$$

holds, where $\mathcal{A}_{\mathsf{H}}$ is a $(q_e, \sigma_e, \mathtt{tm}')$-PRF adversary, $\mathcal{A}_{\Pi}$ is a $(1, \sigma_e, \mathtt{tm}')$-PRIV adversary, $\mathcal{A}'_{\mathsf{H}}$ is a $(q, \sigma, \mathtt{tm}')$-PRF adversary, and $\mathcal{A}'_{\Pi}$ is a $(1, q_d, \sigma_e, \sigma_d, \mathtt{tm}')$-AUTH adversary.

Theorem 10. *Let $\mathcal{A}_{\mathsf{cmt}}$ be a $\mathtt{tm}$-CMT adversary against $\mathsf{HtAE\text{-}ST}[\Pi, \mathsf{H}]$. Then*

$$\mathbf{Adv}^{\mathtt{cmt}}_{\mathsf{HtAE\text{-}ST}[\Pi,\mathsf{H}]}(\mathcal{A}_{\mathsf{cmt}}) \le \mathbf{Adv}^{\mathtt{rcr}}_{\mathsf{H}}(\mathcal{A}_{\mathsf{H}})$$

holds, where $\mathcal{A}_{\mathsf{H}}$ is a $\mathtt{tm}'$-RCR adversary.

Theorem 10 is identical to the original CMT proof [CK25, Theorem 3]. Appendix C shows the proofs of Theorems 9 and 10.

5.9 On CTX and CTY transforms

Despite their use of a mixed model, we do not find instantiation-based attacks against CTX and CTY. Since they are outside our main scope, we briefly outline what could be shown for them. For CTX, let Π be a tag-based AE whose encryption consists of a plain encryption $\Pi.\widetilde{\mathcal{E}} : \mathcal{K} \times \mathcal{N} \times \mathcal{AD} \times \mathcal{M} \to \mathcal{M}$ and a tag generation $\Pi.\mathcal{T} : \mathcal{K} \times \mathcal{N} \times \mathcal{AD} \times \mathcal{M} \to \mathcal{M}$. Encryption of CTX is defined as $\Pi.\mathcal{E}(K, N, A, M) = (\widetilde{C}, \widetilde{T}) = (\Pi.\widetilde{\mathcal{E}}(K, N, A, M), \Pi.\mathcal{T}(K, N, A, M))$. The inverse of $\Pi.\widetilde{\mathcal{E}}$ is denoted by $\Pi.\widetilde{\mathcal{D}} : \mathcal{K} \times \mathcal{N} \times \mathcal{AD} \times \mathcal{M} \to \mathcal{M}$, where $\Pi.\widetilde{\mathcal{D}}(K, N, A, \Pi.\widetilde{\mathcal{E}}(K, N, A, M)) = M$ holds for any valid (N, A, M). See Fig. 12 for their algorithms. First, we can apply KS to CTX and CTY to obtain their standard-model-secure variants. As with (*e.g.*) $\mathsf{AEaH\text{-}KS}$, we use RCR-PRG $\mathsf{G}(K) \to (K_1, K_2)$ for KS, where K_1 is given to Π and K_2 is given to H. We assume H_{K_2} is a CR-PRF. Let $\mathsf{CTX\text{-}KS}[\Pi, \mathsf{G}, \mathsf{H}]$ and $\mathsf{CTY\text{-}KS}[\Pi, \mathsf{G}, \mathsf{H}]$ denote our fixed variants as shown by Fig. 12. We observe that $\mathsf{CTX}[\Pi, \mathsf{H}]$ assuming H as an RO is equivalent to $\mathsf{CTX}[\Pi, \mathsf{R}_{\mathsf{H}}]$ where R_{H} is

$\mathsf{CTX\text{-}KS}.\mathcal{E}[\Pi,\mathsf{G},\mathsf{H}](K,N,A,M)$	$\mathsf{CTX\text{-}KS}.\mathcal{D}[\Pi,\mathsf{G},\mathsf{H}](K,N,A,(\widetilde{C},T))$
1. $(K_1,K_2) \leftarrow \mathsf{G}(K)$ 2. $\widetilde{C} \leftarrow \Pi.\widetilde{\mathcal{E}}(K_1,N,A,M)$ 3. $\widetilde{T} \leftarrow \Pi.\mathcal{T}(K_1,N,A,M)$ 4. $T \leftarrow \mathsf{H}(K_2,N,A,\widetilde{T})$ 5. **return** $(\widetilde{C},T)$	1. $(K_1,K_2) \leftarrow \mathsf{G}(K)$ 2. $M \leftarrow \Pi.\widetilde{\mathcal{D}}(K_1,N,A,\widetilde{C})$ 3. $\widetilde{T}^* \leftarrow \Pi.\mathcal{T}(K_1,N,A,M)$ 4. **if** $T \neq \mathsf{H}(K_2,N,A,\widetilde{T}^*)$: **return** $\perp$ 5. **return** M
$\mathsf{CTY\text{-}KS}.\mathcal{E}[\Pi,\mathsf{G},\mathsf{H}](K,N,A,M)$	$\mathsf{CTY\text{-}KS}.\mathcal{D}[\Pi,\mathsf{G},\mathsf{H}](K,N,A,(\widetilde{C},T))$
1. $(K_1,K_2) \leftarrow \mathsf{G}(K)$ 2. $\widetilde{C} \leftarrow \Pi.\widetilde{\mathcal{E}}(K_1,N,\varepsilon,M)$ 3. $\widetilde{T} \leftarrow \Pi.\mathcal{T}(K_1,N,\varepsilon,M)$ 4. $T \leftarrow \mathsf{H}(K_2,N,A,\widetilde{T})$ 5. **return** $(\widetilde{C},T)$	1. $(K_1,K_2) \leftarrow \mathsf{G}(K)$ 2. $M \leftarrow \Pi.\widetilde{\mathcal{D}}(K_1,N,\varepsilon,\widetilde{C})$ 3. $\widetilde{T}^* \leftarrow \Pi.\mathcal{T}(K_1,N,\varepsilon,M)$ 4. **if** $T \neq \mathsf{H}(K_2,N,A,\widetilde{T}^*)$: **return** $\perp$ 5. **return** M

Fig. 12: Standard-model secure versions of CTX and CTY transforms. (Top) CTX-KS, (Bottom) CTY-KS. The original $\mathsf{CTX}[\Pi,\mathsf{H}]$ and $\mathsf{CTY}[\Pi,\mathsf{H}]$ are obtained by omitting the line $[(K_1,K_2) \leftarrow \mathsf{G}(K)]$ and using K instead of K_1 and K_2.

a URF compatible with H if the adversary has no primitive queries to H inside $\mathsf{CTX}[\Pi,\mathsf{H}]$. Therefore, AE security bound for $\mathsf{CTX\text{-}KS}[\Pi,\mathsf{G},\mathsf{H}]$ will be obtained as a sum of PRG and PRF terms, and that obtained by the original bound of [CR22] with the number of primitive queries set to zero.[10] For CMT security, the proof and the bound, the sum of RCR bound of G and CR bound of H, are mostly identical to those of AEaH-KS (Theorem 2). The same analysis applies to CTY: the AE security bound of $\mathsf{CTY\text{-}KS}[\Pi,\mathsf{G},\mathsf{H}]$ is obtained by the sum of PRG and PRF terms and the original bound of (RO-based) CTY [BH24] with omission of primitive queries. CMT security bound is similarly obtained as CTX-KS.

Dedicated analyses might yield better, finer AE security bounds, and there may be options beyond KS for achieving standard model security. In particular, proving/disproving the standard-model security of CTX/CTY without changing the algorithm, as we did for HtAE, remains open. They would be interesting future topics.

6 Conclusions

We have revisited state-of-the-art generic transforms that rely on a mixed security model. Although we have not invalidated the original proofs, we have demonstrated practical counterexamples against them once instantiated. An idealized model is useful and a mixed model is logical and can provide efficient solutions for many constructions. However, considering the principal purpose of generic transforms, our results warn against using them without applying adequate fixes, including ours. Put differently, our work suggests that additional

[10] This observation in principle holds for CK25 transforms that we fixed; however, we provided their proofs from scratch as they are in our main scope, and for completeness and cleanness. Consequently, we simplified some proofs and improved some bounds relative to those obtained by a direct application of CK25 bounds.

instantiation-specific care must be taken when using generic transforms that rely on a mixed model in real-world applications.

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A Proofs of Theorems 5 and 6 (EtH-KS)

A.1 Proof of Theorem 5

As with other proofs, we skip the hybrids regarding the replacement of G and H. Let Θ_1 denote $\mathsf{EtH\text{-}KS}[\Gamma, \$_{\mathsf{G}}, \mathsf{R}_{\mathsf{H}}]$. We have

$$\begin{aligned}\mathbf{Adv}^{\mathtt{priv}}_{\mathsf{EtH\text{-}KS}[\Gamma,\mathsf{G},\mathsf{H}]}(\mathcal{A}_{\mathsf{priv}}) &\le \mathbf{Adv}^{\mathtt{ind}}_{\mathsf{EtH\text{-}KS}[\Gamma,\mathsf{G},\mathsf{H}].\mathcal{E},\Theta_1.\mathcal{E}}(\mathcal{A}_{\mathsf{priv}}) + \mathbf{Adv}^{\mathtt{ind}}_{\Theta_1.\mathcal{E},\$}(\mathcal{A}_{\mathsf{priv}}) \\ &\le \mathbf{Adv}^{\mathtt{prg}}_{\mathsf{G}}(\mathcal{A}_{\mathsf{G}}) + \mathbf{Adv}^{\mathtt{prf}}_{\mathsf{H}}(\mathcal{A}_{\mathsf{H}}) + \mathbf{Adv}^{\mathtt{priv}}_{\Gamma}(\mathcal{A}_{\Gamma}).\end{aligned}$$

As in the case of the privacy bound proof of $\mathsf{AEaH\text{-}KS}$, $\mathbf{Adv}^{\mathtt{ind}}_{\Theta_1.\mathcal{E},\$}(\mathcal{A}_{\mathsf{priv}}) \le \mathbf{Adv}^{\mathtt{priv}}_{\Gamma}(\mathcal{A}_{\Gamma})$ holds as $\mathcal{A}_{\mathsf{priv}}$ is nonce-respecting and the T part is always independently uniform and random for both worlds. For authenticity bound, we have

$$\begin{aligned}&\mathbf{Adv}^{\mathtt{auth}}_{\mathsf{EtH\text{-}KS}[\Gamma,\mathsf{G},\mathsf{H}]}(\mathcal{A}_{\mathsf{auth}}) \\ &\le \mathbf{Adv}^{\mathtt{ind}}_{\mathsf{EtH\text{-}KS}[\Gamma,\mathsf{G},\mathsf{H}],\Theta_1}(\mathcal{A}'_{\mathsf{auth}}) + \mathbf{Adv}^{\mathtt{auth}}_{\Theta_1}(\mathcal{A}_{\mathsf{auth}}) \\ &\le \mathbf{Adv}^{\mathtt{prg}}_{\mathsf{G}}(\mathcal{A}'_{\mathsf{G}}) + \mathbf{Adv}^{\mathtt{prf}}_{\mathsf{H}}(\mathcal{A}'_{\mathsf{H}}) + \mathbf{Adv}^{\mathtt{auth}}_{\Theta_1}(\mathcal{A}_{\mathsf{auth}}).\end{aligned}$$

We then have $\mathbf{Adv}^{\mathtt{auth}}_{\Theta_1}(\mathcal{A}_{\mathsf{auth}}) \le q_d/2^t$ since, for a decryption query (N', A', C', T'), the probability that $\mathsf{R}_{\mathsf{H}}(K_2, N', A', C') = T'$ holds is at most $1/2^t$, which means the probability that $\mathcal{A}_{\mathsf{auth}}$ obtains some outputs other than $\perp$ from $\Theta_1.\mathcal{D}$ in the game is at most $q_d/2^t$. This concludes the proof.

A.2 Proof of Theorem 6

Let $Z = ((K, N, A, M), (K', N', A', M'), (C, T))$ be a tuple. Let $(K_1, K_2) = \mathsf{G}(K)$ and $(K'_1, K'_2) = \mathsf{G}(K')$. When $K = K'$, $(N, A) = (N', A')$ implies $M = M'$, thus Z is not valid (the attack never succeeds). If $K = K'$ and $(N, A) \neq (N', A')$ hold, it implies H's collision. When $K \neq K'$, if $K_2 \neq K'_2$ occurs it implies a collision of H, and if $K_2 = K'_2$ occurs it implies right-half collision of G. Thus, a valid Z either requires a collision on H or a right-half collision of G. This concludes the proof.

B Proofs of Theorems 7 and 8 (chaSIV-NKS)

B.1 Proof of Theorem 7

The target scheme is $\mathsf{chaSIV\text{-}NKS}[\Gamma, \mathsf{F}, \mathsf{H}]$. For proving the MR-PRIV bound, we first apply the hybrid argument to the original by replacing F with a URF R_F. This incurs additional $\mathbf{Adv}^{\mathtt{prf}}_{\mathsf{F}}(\mathcal{A}_\mathsf{F})$ for $\mathcal{A}_\mathsf{F}$ using $2Q_e$ queries. After the hybrid argument, the scheme is transformed into the composition of Γ and H as in the original chaSIV. The difference is that, for each query taking nonce N, the scheme first uniformly samples $(K_e^N, K_h^N) \leftarrow\!\$\, \mathcal{K}^2$ independently for each N, and Γ takes the (nonce-dependent) key K_e^N. The PRF H takes (A, M) as input with the key K_h^N. We define this scheme as $\mathsf{chaSIV\text{-}NKS'}[\Gamma, \mathsf{H}]$ and denote it by Θ_1. Moreover, let Θ_2 denote $\mathsf{chaSIV\text{-}NKS'}[\Pi, \mathsf{R}_{\widetilde{\mathsf{H}}}]$ which computes $T = \mathsf{R}_{\widetilde{\mathsf{H}}}(N, A, M)$ for a URF $\mathsf{R}_{\widetilde{\mathsf{H}}}$. The rest is identical to Θ_1; Θ_2 ignores K_h^N and randomly generates K_e^N per N and uses it for Γ.

For the MR-PRIV bound, we have

$$\mathbf{Adv}^{\mathtt{mr\text{-}priv}}_{\mathsf{chaSIV\text{-}NKS}[\Gamma,\mathsf{F},\mathsf{H}]}(\mathcal{A}_{\mathsf{priv}}) \leq \mathbf{Adv}^{\mathtt{mr\text{-}priv}}_{\mathsf{chaSIV\text{-}NKS'}[\Gamma,\mathsf{H}]}(\mathcal{A}_{\mathsf{priv}}) + \mathbf{Adv}^{\mathtt{prf}}_{\mathsf{F}}(\mathcal{A}_\mathsf{F}), \quad (6)$$

$$\mathbf{Adv}^{\mathtt{mr\text{-}priv}}_{\mathsf{chaSIV\text{-}NKS'}[\Gamma,\mathsf{H}]}(\mathcal{A}_{\mathsf{priv}}) \leq \mathbf{Adv}^{\mathtt{ind}}_{\Theta_1.\mathcal{E},\Theta_2.\mathcal{E}}(\mathcal{A}_{\mathsf{priv}}) + \mathbf{Adv}^{\mathtt{ind}}_{\Theta_2.\mathcal{E},\$}(\mathcal{A}_{\mathsf{priv}}),$$

$$\mathbf{Adv}^{\mathtt{ind}}_{\Theta_1.\mathcal{E},\Theta_2.\mathcal{E}}(\mathcal{A}_{\mathsf{priv}}) \leq Q_e \cdot \mathbf{Adv}^{\mathtt{prf}}_{\mathsf{H}}(\mathcal{A}_\mathsf{H}), \quad (7)$$

$$\mathbf{Adv}^{\mathtt{ind}}_{\Theta_2.\mathcal{E},\$}(\mathcal{A}_{\mathsf{priv}}) \leq Q_e \cdot \mathbf{Adv}^{\mathtt{priv}}_{\Gamma}(\mathcal{A}_\Gamma) + \frac{\sum_{i\in[1..Q_e]} R_i^2}{2^r} \quad (8)$$

holds. Eq. (7) holds from a simple multi-instance analysis of PRF: there are Q_e instances for H each taking at most R_e^* queries in Θ_1, while $\mathsf{R}_{\widetilde{\mathsf{H}}}$ takes N as a part of input. To see Eq. (8) holds, we observe that the T part is uniform for both worlds, thus the distinguishing game reduces to the multi-instance distinguishing game for Γ, where each instance is indexed by N, and there are Q_e instances. The bad event is a collision of $N_e = \mathtt{msb}_r(T)$ to Γ under the same instance, *i.e.*, under the same K_e^N, in Θ_2. Since T is completely random, the bad event occurs with probability at most $\sum_{i\in[1..Q_e]} R_i^2/2^r$. If this bad event does not occur, the game reduces to the standard multi-instance PRIV game for Γ under the condition that N_es do not collide among the same instance. The advantage of this game is at most $Q_e \cdot \mathbf{Adv}^{\mathtt{priv}}_{\Gamma}(\mathcal{A}_\Gamma)$, as the PRIV game allows arbitrarily distinct choice of N_e. From Eqs. (6) to (8), we conclude the MR-PRIV proof.

For the MR-AUTH bound, we define Θ_1 and Θ_2 as in the MR-PRIV proof. Thus, we have

$$\begin{aligned}
\mathbf{Adv}^{\mathtt{mr\text{-}auth}}_{\mathsf{chaSIV\text{-}NKS}[\Gamma,\mathsf{F},\mathsf{H}]}(\mathcal{A}_{\mathsf{auth}}) &\le \mathbf{Adv}^{\mathtt{mr\text{-}auth}}_{\mathsf{chaSIV\text{-}NKS}'[\Gamma,\mathsf{H}]}(\mathcal{A}_{\mathsf{auth}}) + \mathbf{Adv}^{\mathtt{prf}}_{\mathsf{F}}(\mathcal{A}'_{\mathsf{F}}),\\
\mathbf{Adv}^{\mathtt{mr\text{-}auth}}_{\mathsf{chaSIV\text{-}NKS}'[\Gamma,\mathsf{H}]}(\mathcal{A}_{\mathsf{auth}}) &\le \mathbf{Adv}^{\mathtt{ind}}_{\Theta_1,\Theta_2}(\mathcal{A}'_{\mathsf{auth}}) + \mathbf{Adv}^{\mathtt{auth}}_{\Theta_2}(\mathcal{A}_{\mathsf{auth}}). \qquad (9)
\end{aligned}$$

The term $\mathbf{Adv}^{\mathtt{ind}}_{\Theta_1,\Theta_2}(\mathcal{A}'_{\mathsf{auth}})$ in Eq. (9) is at most $Q_{ed}\cdot\mathbf{Adv}^{\mathtt{prf}}_{\mathsf{H}}(\mathcal{A}'_{\mathsf{H}})$. For the last term in Eq. (9), we relax the game so that K_e^N is given to the adversary on any query involving N. The tidiness assumption on Γ ensures that (K_e^N, C) for a valid ciphertext C uniquely determines the corresponding plaintext M. Thanks to the tidiness, the game effectively reduces to the deterministic MAC unforgeability game for $\mathsf{R}^{\sim}_{\mathsf{H}}$ on the message (N, A, M); note that N is part of the message, hence it can repeat. Since $\mathsf{R}^{\sim}_{\mathsf{H}}$ is a URF, the forgery probability (which is equal to the last term of Eq. (9)) is $q_d/2^t$. This proves MR-AUTH bound.

B.2 Proof of Theorem 8

The proof is mostly identical to the proof of Theorem 2. Let Z denote a tuple consisting of (K, N, A, M), (K', N', A', M') and (C, T). Let $(K_e, K_h) = \mathsf{F}(K, N)$ and $(K'_e, K'_h) = \mathsf{F}(K', N')$. If $(N, K) = (N', K')$, we have $(K_e, K_h) = (K'_e, K'_h)$, which implies $M = M'$ and $A \neq A'$. This implies a collision on H. If $(N, K) \neq (N', K')$, the subcase $(K_h = K'_h)$ implies a right-half collision on F. Another subcase $(K_h \neq K'_h)$ implies a collision on H. Thus, a valid Z implies a collision on H or a right-half collision on F, which concludes the proof.

C Proofs of Theorems 9 and 10 (HtAE-ST)

C.1 Proof of Theorem 9

As with other proofs, let Θ_1 denote $\mathsf{HtAE\text{-}ST}[\Pi, \mathsf{R}_{\mathsf{H}}]$. For PRIV bound, we have

$$\begin{aligned}
\mathbf{Adv}^{\mathtt{priv}}_{\mathsf{HtAE\text{-}ST}[\Pi,\mathsf{H}]}(\mathcal{A}_{\mathsf{priv}}) &\le \mathbf{Adv}^{\mathtt{ind}}_{\mathsf{HtAE\text{-}ST}[\Pi,\mathsf{H}].\mathcal{E},\Theta_1.\mathcal{E}}(\mathcal{A}_{\mathsf{priv}}) + \mathbf{Adv}^{\mathtt{ind}}_{\Theta_1.\mathcal{E},\$}(\mathcal{A}_{\mathsf{priv}})\\
&\le \mathbf{Adv}^{\mathtt{prf}}_{\mathsf{H}}(\mathcal{A}_{\mathsf{H}}) + q_e\mathbf{Adv}^{\mathtt{priv}}_{\Pi}(\mathcal{A}_{\Pi}),
\end{aligned}$$

where $\mathcal{A}_\Pi$ is a PRIV adversary against Π involving a randomly chosen key, and it performs one encryption query. $\mathbf{Adv}^{\mathtt{ind}}_{\Theta_1.\mathcal{E},\$}(\mathcal{A}_{\mathsf{priv}}) \le q_e\mathbf{Adv}^{\mathtt{priv}}_{\Pi}(\mathcal{A}_{\Pi})$ holds as $\mathcal{A}_{\mathsf{priv}}$ is nonce-respecting, and the T part is always independently uniform and random for both worlds; moreover, a secret key of Π is chosen uniformly at random every time $\mathcal{A}_{\mathsf{priv}}$ performs an encryption query.

For authenticity, let $\Pi_\perp = (\Pi.\mathcal{E}, \perp)$ and $\Theta_2 = \mathsf{HtAE\text{-}ST}[\Pi_\perp, \mathsf{R}_{\mathsf{H}}]$. We have

$$\begin{aligned}
\mathbf{Adv}^{\mathtt{auth}}_{\mathsf{HtAE\text{-}ST}[\Pi,\mathsf{H}]}(\mathcal{A}_{\mathsf{auth}}) &\le \mathbf{Adv}^{\mathtt{prf}}_{\mathsf{H}}(\mathcal{A}'_{\mathsf{H}}) + \mathbf{Adv}^{\mathtt{auth}}_{\Theta_1}(\mathcal{A}_{\mathsf{auth}})\\
&\le \mathbf{Adv}^{\mathtt{prf}}_{\mathsf{H}}(\mathcal{A}'_{\mathsf{H}}) + \mathbf{Adv}^{\mathtt{ind}}_{\Theta_1,\Theta_2}(\mathcal{A}'_{\mathsf{auth}}) + \mathbf{Adv}^{\mathtt{auth}}_{\Theta_2}(\mathcal{A}_{\mathsf{auth}})\\
&\le \mathbf{Adv}^{\mathtt{prf}}_{\mathsf{H}}(\mathcal{A}'_{\mathsf{H}}) + Q_d\cdot\mathbf{Adv}^{\mathtt{auth}}_{\Pi}(\mathcal{A}'_{\Pi}) + \frac{q_d}{2^t},
\end{aligned}$$

where $\mathcal{A}'_{\Pi}$ is an AUTH adversary against Π involving a randomly chosen key, and it performs at most one encryption query. In the above evaluations, we have $\mathbf{Adv}^{\mathtt{ind}}_{\Theta_1,\Theta_2}(\mathcal{A}'_{\mathsf{auth}}) \leq Q_d \mathbf{Adv}^{\mathtt{auth}}_{\Pi}(\mathcal{A}'_{\Pi})$ similarly to the PRIV case; observe that there may be instances (indexed by N) taking no decryption queries, however, they do not contribute to the advantage by definition. Thus, only Q_d instances can contribute to the advantage. The reason why $\mathbf{Adv}^{\mathtt{auth}}_{\Theta_2}(\mathcal{A}_{\mathsf{auth}}) \coloneqq \mathbf{Adv}^{\mathtt{ind}}_{\Theta_2,(\Theta_2.\mathcal{E},\perp)}(\mathcal{A}_{\mathsf{auth}}) \leq q_d/2^t$ holds is as follows. We say the tuple (N', A', T') in a decryption query (N', A', C', T') is *new* if $\mathcal{A}_{\mathsf{auth}}$ has not queried (N', A', M) to $\Theta_2.\mathcal{E}$ and received (C, T') from $\Theta_2.\mathcal{E}$ for some M and C. When a decryption query (N', A', C', T') contains a new tuple (N', A', T'), the distinguishing probability between Θ_2 and $(\Theta_2.\mathcal{E}, \perp)$ is at most $1/2^t$ as R_{H} is a random function. When (N', A', T') is not new, there exists an encryption query and its response $(N', A', M, (C, T'))$ and $C' \neq C$ holds. Thus, the invoked decryption query to $\Pi_{\perp}.\mathcal{D}$, i.e. (N', A', C'), is not a forbidden query in the authenticity game against $\Pi_{\perp}$, and $\Pi_{\perp}.\mathcal{D}$ will return $\perp$ for that query. Therefore, the distinguishing probability between Θ_2 and $(\Theta_2.\mathcal{E}, \perp)$ is zero with such a query. From the above discussions, we have $\mathbf{Adv}^{\mathtt{auth}}_{\Theta_2}(\mathcal{A}_{\mathsf{auth}}) \leq q_d/2^t$.

C.2 Proof of Theorem 10

Suppose $Z = ((K, N, A, M), (K', N', A', M'), (C, T))$ be the final output of $\mathcal{A}_{\mathsf{cmt}}$ with the corresponding tuple (C, T). For any (K, N, A), let $\mathsf{H}^{\mathrm{left}}(K, N, A) = K_e$ where $\mathsf{H}(K, N, A) = (K_e, T)$. Let $K_e = \mathsf{H}^{\mathrm{left}}(K, N, A)$ and $K'_e = \mathsf{H}^{\mathrm{left}}(K', N', A')$. When $(K, N, A) = (K', N', A')$, we have $K_e = K'_e$ and $M = M'$ since

$$\Pi.\mathcal{D}(K_e, N, \varepsilon, C) = \Pi.\mathcal{D}(K'_e, N', \varepsilon, C)$$

holds, which means Z is not valid. When $(K, N, A) \neq (K', N', A')$, the probability that $\mathsf{H}^{\mathrm{right}}(K, N, A) = \mathsf{H}^{\mathrm{right}}(K', N', A')$ holds is equal to $\mathbf{Adv}^{\mathtt{rcr}}_{\mathsf{H}}(\mathcal{A}_{\mathsf{H}})$ for some $\mathcal{A}_{\mathsf{H}}$ by definition. Therefore, we have

$$\mathbf{Adv}^{\mathtt{cmt}}_{\mathsf{HtAE\text{-}ST}[\Pi,\mathsf{H}]}(\mathcal{A}_{\mathsf{cmt}}) \leq \mathbf{Adv}^{\mathtt{rcr}}_{\mathsf{H}}(\mathcal{A}_{\mathsf{H}}),$$

which concludes the proof.